Constraining the Distribution of 3D Fractal Structures in Mud Flocs

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Key Points:

- Aggregating floc data overestimates 3D fractal dimension due to structural diversity.
- Stratifying by 2D box-counting dimension reveals trends with shear velocity and organic matter.
- Revised fractal dimensions indicate lower settling rates than conventional aggregated estimates.

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Abstract

Mud builds coastal landscapes and carries pollutants and carbon, yet predicting its transport is difficult because mud aggregates into flocs that control its settling rate in river, estuarine, and marine settings. Flocs are typically characterized using fractal theory with a constant fractal dimension ($n_f = 2.0$), which is inferred from aggregating floc size–settling velocity data, despite a diversity of complex fractal structures. We conducted experiments on freshwater flocs under varied shear stress and organic concentration to characterize this diversity. Rather than the conventional approach of aggregating settling data to infer n_f, we grouped measurements by the two-dimensional (2D) box-counting dimension derived from particle tracking, thereby partially resolving structural variability. This stratified method reveals that n_f increases with shear velocity and decreases with organic matter content, trends that are obscured when data are aggregated. The stratified analysis resolves n_f variations of 0.2–0.35 units across experimental conditions, whereas the conventional aggregated approach reduces this range by roughly a factor of five. The inferred primary particle diameter d_p co-varies with these conditions, and the concurrent variation of n_f, d_p, and floc size introduces competing effects on settling velocity. Our approach yields lower n_f values (1.6–2.1) than the canonical range, which propagates nonlinearly into settling rates, implying that mud flocs are more porous and settle more slowly than expected. We also find that the 2D-to-3D n_f relationship can be described by a nonlinear model, consistent with empirical conversions for synthetic aggregates. Overall, our results indicate that mud retention in floodplains and deltas is less efficient than current floc models assume, at least in the freshwater conditions we analyzed, with implications for sediment budgets, contaminant dispersal, and coastal landscape evolution.

Plain Language Summary

Mud particles in rivers tend to clump together into clusters called flocs. How quickly these flocs sink determines where mud is deposited, shaping the growth of river deltas, floodplains, and coastlines. Predicting this settling is challenging because flocs vary widely in structure: some are dense and compact, while others are porous and loosely bound. Most studies assume that all flocs have a single, uniform structure, which oversimplifies their behavior. In our experiments, we generated flocs in freshwater under controlled conditions with different flow strengths and amounts of organic matter. Using high-resolution imaging, we measured both their structure and settling speed. We found that faster flows tend to produce smaller but more compact flocs, while organic matter encourages the formation of more open structures that settle more slowly. By grouping flocs according to their measured structural complexity, we revealed trends that are hidden when data are averaged together. This new approach reduces bias in estimating settling rates and improves predictions of how mud moves through rivers and into estuaries and oceans, where salinity may further modify aggregation beyond the freshwater trends isolated here.

1 Introduction

Fine-grained sediment, or mud, defined by particles smaller than 62.5 µm, dominates fluvial sediment loads and represents the dominant material deposited in lowland and deltaic environments (Healy et al., 2002; Kranenburg, 1994; Lamb et al., 2020; Nghiem et al., 2026). The dynamics of mud transport and deposition therefore exert a first-order control on the long-term evolution of deltas, floodplains, and estuaries (Nghiem et al., 2022; Winterwerp, 2002), as well as on water quality, contaminant fluxes, and the success of restoration interventions (Droppo et al., 1997; Esposito et al., 2017; Kemp et al., 2016; Liss et al., 1996). Mud also plays a central role in the global carbon cycle by providing reactive mineral surfaces for organic carbon adsorption and long-term preservation (Bianchi et al., 2024; Blair & Aller, 2012).

The geomorphic and geochemical importance of mud necessitates a mechanistic framework for particle transport and settling to enable predictive modeling. Settling velocity is a key parameter in sediment-transport equations and is commonly estimated using Stokes' law, which gives the terminal settling velocity (w_s) of smooth, impermeable spheres at low Reynolds numbers (Bird, 2006; Stokes et al., 1851):

$$w_s = \frac{g\,(\rho_p - \rho_f)\,d^2}{18\,\rho_f\,\nu} \tag{1}$$

where $g = 9.81\ \mathrm{m\,s^{-2}}$ is gravitational acceleration, ρ_p ($\mathrm{kg\,m^{-3}}$) and ρ_f ($\mathrm{kg\,m^{-3}}$) are the densities of the particle and fluid, d (m) is particle diameter, and ν ($\mathrm{m^2\,s^{-1}}$) is the kinematic viscosity of the fluid.

However, when applied to fine-grained sediment, Stokes' law yields predictions at odds with field observations. Consider a single 1 µm mud particle in a 5 m-deep river flowing at 1 $\mathrm{m\,s^{-1}}$. Using Stokes' law produces a settling velocity of just $9\times10^{-8}\ \mathrm{m\,s^{-1}}$, implying a deposition time greater than 600 days and a transport distance exceeding 55,000 km downstream. This hypothetical scenario conflicts sharply with field observations of mud deposition in fluvial and coastal settings (Lamb et al., 2020; Mason & Mohrig, 2019; Nicholas & Walling, 1996).

This discrepancy indicates that mud settles far more rapidly than predicted for isolated particles, pointing to additional physical processes. The deviation arises because suspended mud particles flocculate, forming larger aggregates composed of individual primary particles that settle much faster than their individual constituents (Droppo et al., 1997; Johnson et al., 1996; Kranenburg, 1994). These flocculated aggregates, or flocs, are highly porous rather than dense, impermeable spheres, so their effective density decreases with increasing size (Kranenburg, 1994; Meakin, 1992). Because flocs exhibit approximately self-similar fractal organization (Family & Landau, 2012; Jullien, 1987), their hierarchical structure can be described by a fractal mass–size scaling (Meakin, 1992):

$$N \sim \left(\frac{d_f}{d_p}\right)^{n_f}, \tag{2}$$

where N is the number of primary particles (dimensionless), d_f is the floc diameter (m), d_p is the primary particle diameter (m), and n_f is the fractal dimension (dimensionless).

Building on Eq. (2), the fraction of the floc volume occupied by solid grains is the ratio of solid volume to the enclosing floc volume. Because the solid volume scales with N times the monomer volume while the enclosure scales with d_f^3, the solid volume fraction ϕ decreases with size approximately as $(d_f/d_p)^{n_f-3}$. For a porous aggregate whose pores are filled by ambient water, the floc's submerged specific gravity equals this solid fraction times the solid's submerged specific gravity. Denoting submerged specific gravity by $R = (\rho - \rho_w)/\rho_w$, this yields:

$$R_f = R_s \left(\frac{d_f}{d_p}\right)^{n_f-3}, \tag{3}$$

where R_f and R_s are the submerged specific gravity of the floc and primary particles, respectively (both dimensionless), ρ is particle density ($\mathrm{kg\,m^{-3}}$), and ρ_w is water density ($\mathrm{kg\,m^{-3}}$). Since a dense solid sphere has volume that scales with the cube of its diameter, its fractal dimension is $n_f = 3$, and Eq. (3) reduces to $R_f = R_s$. Naturally formed flocs are more porous with $1 \le n_f < 3$, yielding $R_f < R_s$. Accordingly, floc effective (excess) density decreases with floc size. For a fractal aggregate, the solids mass scales as $M_f \propto d_f^{n_f}$ whereas the envelope volume scales as $V_f \propto d_f^3$, implying $\Delta\rho_f \propto M_f/V_f \propto d_f^{n_f-3}$ (Kranenburg, 1994).

Substituting Eq. (3) into Eq. (1) gives an explicit expression for the settling velocity of a fractal floc (Strom & Keyvani, 2011; Winterwerp, 1998):

$$w_s = \frac{g\,R_s}{b_1\,\nu\,d_p^{n_f-3}}\,d_f^{n_f-1}, \tag{4}$$

where w_s is the settling velocity ($\mathrm{m\,s^{-1}}$), g is gravitational acceleration ($\mathrm{m\,s^{-2}}$), ν is kinematic viscosity ($\mathrm{m^2\,s^{-1}}$), and b_1 is a dimensionless shape factor (with $b_1 = 18$ for smooth spheres).

In practice, applying Eq. (4) requires an appropriate fractal dimension n_f, which captures how floc structure modifies settling velocity. Experimental and field studies often assume $n_f \approx 2$ from empirical w_s–d_f relationships, values reported primarily for saline estuarine/coastal flocs where electrochemical interactions promote more compact aggregates; freshwater flocs more often show lower n_f ($\lesssim 2$), with overlap depending on turbulence and composition (Winterwerp, 1998; J. Winterwerp et al., 2006). With all other parameters held fixed, Eq. (4) gives $w_s \propto d_f^{n_f-1}$ (Johnson et al., 1996; Li & Ganczarczyk, 1989). Measurements of w_s and d_f across a floc population therefore allow n_f to be inferred from the log–log slope of the w_s–d_f relation (Kranenburg, 1994; Strom & Keyvani, 2011; Winterwerp, 1998, 2002; J. C. Winterwerp & Van Kesteren, 2004; J. Winterwerp et al., 2006). In general, this inference assumes that floc permeability is independent of floc size. If permeability instead varies systematically with d_f, the observed scaling relation may reflect both fractal structure and permeability effects (Strom & Keyvani, 2011; Nghiem et al., 2026).

Using a constant population-wide value for the fractal dimension is convenient but restrictive. Khelifa and Hill (2006) showed that treating n_f as constant fails to reproduce the observed spread in settling velocities and argued that n_f can decrease with floc size, motivating models that prescribe a function $n_f(d_f)$, often as a power law, to represent within-population variability. In practice, however, most experimental and image-based field studies measure only floc size d_f and settling velocity w_s (Smith & Friedrichs, 2015; Nghiem et al., 2024), so the fractal dimension is not independently observed and functions for n_f cannot be directly tested. Moreover, models for $n_f(d_f)$ implicitly assume a unique fractal dimension for a given floc size (Khelifa & Hill, 2006), whereas natural floc populations can contain multiple flocs with comparable d_f but distinct internal structure, and therefore distinct n_f (Maggi, 2007). As a result, standard log–log regressions of w_s on d_f treat structurally heterogeneous populations as homogeneous, and pooling such subpopulations can misattribute between-group differences to size scaling itself, producing biased estimates of n_f.

To illustrate this effect, consider a population comprising three subgroups of flocs, each defined by a distinct fractal dimension (n_f = 1.5, 2.0, and 2.5) but formed from primary particles of the same diameter ($d_p = 10^{-5}$ m). According to Eq. (4), the slope of the log–log relationship between settling velocity and floc diameter for each subgroup is $n_f - 1$. When analyzed separately, these subgroups yield distinct log–log slopes corresponding to n_f = 1.5, 2.0, and 2.5, respectively (Fig. 1). However, if these floc subgroups are combined into a single group and floc diameter (d_f) is correlated with the 3D fractal dimension (n_f), then fitting a single power-law relationship to all points can yield a substantially skewed and misleading exponent. In Fig. 1, the resulting log–log slope is 2, implying $n_f = 3$, which corresponds to the settling behavior of smooth, impermeable spheres, not flocs. This example parallels Simpson's Paradox (Simpson, 1951), a statistical phenomenon in which relationships observed within individual subgroups are reversed, masked, or distorted when data are pooled. In the context of floc settling, pooling structurally distinct subpopulations can produce a global w_s–d_f slope that implies an apparent n_f inconsistent with the n_f of any subgroup, so the inferred "representative" fractal dimension is a regression artifact rather than a physical descriptor.

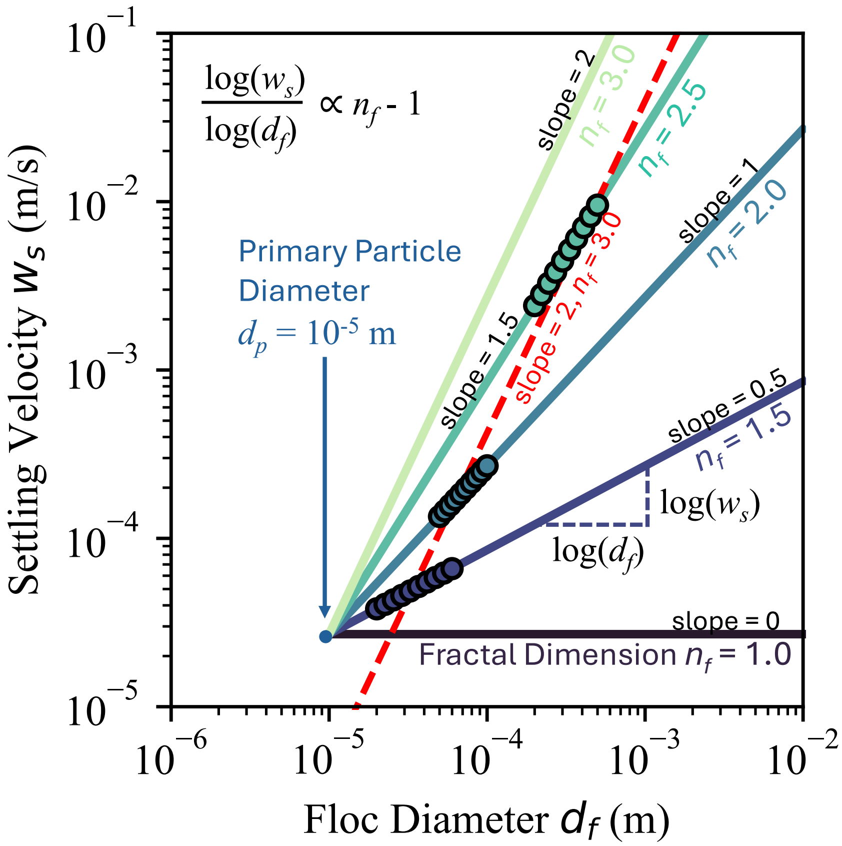

Figure 1. Synthetic data illustrating Simpson's Paradox in floc settling. Three floc subgroups, each with a specific 3D fractal dimension ($n_f = 1.5, 2.0$, and 2.5), are shown. By Eq. (4), the slope of the log–log relationship between settling velocity (w_s) and floc diameter (d_f) is $n_f - 1$. Fitting each subgroup individually recovers slopes of 0.5, 1.0, and 1.5, as expected. In contrast, pooling all points produces a single power-law fit (red line) with a misleading slope of 2, equivalent to $n_f = 3$, characteristic of solid spheres and not flocs. All trend lines intersect at the primary particle diameter ($d_f = d_p = 10^{-5}$ m), where settling velocity is independent of fractal dimension.

To estimate fractal dimension independently of settling velocity and floc diameter, studies have employed image-based techniques that analyze in situ 2D projections of flocs (Jarvis et al., 2005). These approaches typically calculate a two-dimensional (2D) fractal dimension from high-resolution images, then infer the corresponding three-dimensional (3D) structure via empirical or simulation-based relationships. A common method relates the perimeter–area scaling of projected flocs to a 2D fractal dimension, using computationally generated aggregates to derive a mapping between 2D and 3D fractal dimension (Lee & Kramer, 2004; Maggi & Winterwerp, 2004; Tang & Maggi, 2015). However, the relationship between 2D and 3D fractal dimension is derived from synthetic aggregates, which may not replicate the aggregation dynamics or structural variability of natural flocs. Projection-based estimates therefore risk systematic bias and, more importantly, do not recover the 3D fractal dimension that directly controls settling behavior.

Although flocculation has been widely characterized, how within-population structural heterogeneity affects settling remains uncertain, particularly under natural particle-size distributions and aggregation pathways. The central difficulty is that w_s–d_f regressions alone do not condition on structure, so pooling distinct subpopulations can produce an apparent relation and inferred n_f that is biased by data-pooling (Simpson-type) effects. To address this bias, we use images from experiments to stratify the floc population by the 2D box-counting dimension and within each subgroup, regress settling velocity w_s against floc diameter d_f to estimate the hydrodynamically relevant 3D fractal dimension n_f. This conditioning step provides a principled way to separate structural heterogeneity from size scaling without prescribing a population-level closure such as $n_f(d_f)$: that is, n_f is inferred hydrodynamically within approximately homogeneous structural classes. The assumption is that flocs with similar 2D box-counting dimension have similar 3D fractal dimension, which we validate herein.

We evaluate this framework in controlled laboratory experiments designed to simulate flocs in rivers with systematic variation in shear velocity and organic matter concentration. High-resolution in situ imaging enabled quantification of floc morphology and settling trajectories. Specifically, we pursued three objectives: (1) demonstrate the existence of structurally distinct subgroups within the floc population, each characterized by a unique fractal dimension; (2) relate 2D image–derived fractal dimensions to settling-inferred 3D dimensions, and assess this relationship against theoretical predictions; and (3) estimate the effective primary particle size (d_p) and quantify how the hydrodynamic shape factor (b_1) varies as a function of n_f. All experiments were conducted in freshwater (salinity ≈ 0), so the results isolate the roles of shear and organic binding and are not intended as a direct constraint for saline estuaries where salt-driven cohesion can further modify flocculation.

2 Methodology

2.1 Experimental Setup

We performed flocculation experiments in a custom mixing tank designed to provide controlled conditions (Fig. 2a). The tank comprises a cylindrical PVC pipe, 20 cm in diameter and 60 cm tall, filled with water to a depth of 40 cm. The tank was filled with fresh tap water, used without filtration or chemical modification, and experiments were conducted at room temperature. A motor-driven rotating lid, mounted above the open top and submerged just below the free surface, sets the desired shear. For each experiment, the tank was filled with water and the motor was started to establish the target shear. Particles and organic matter (described below) were then added gradually, and the tank was mixed for 30 min to allow flocs to form under sustained shear. The motor was subsequently stopped to initiate settling, and video recording began immediately and continued for the first 30 min.

Shear velocity, u_*, was empirically calibrated for each run by converting motor revolutions per minute to u_* using a relationship derived from previous sediment-entrainment trials in Douglas et al. (2025). We define $u_* = (\tau/\rho_w)^{1/2}$, where τ is a characteristic bed shear stress, and we treat u_* as a characteristic measure of turbulent shear intensity because the rotating lid-driven flow field and boundary stresses are spatially heterogeneous.

During the mixing phase, the flow was fully turbulent. This is quantified by the friction Reynolds number $Re_\tau = u_* h/\nu$, where h is the water depth and ν is the kinematic viscosity of freshwater. For $h = 0.40$ m and $\nu \approx 1.0{\times}10^{-6}$ m^2 s^{-1} at room temperature, the imposed shear velocities $u_* = 0.02$–0.04 m s^{-1} correspond to $Re_\tau \approx (8\text{–}16)\times 10^3$, well within the fully turbulent regime for wall-bounded shear flows (Pope, 2000; Schlichting & Gersten, 2000). Across the flocculation experiments, we imposed $u_* = 0.02$–0.04 m s^{-1} (equivalent to $\tau \approx 0.4$–1.6 Pa), spanning low-to-moderate shear conditions typical of suspension-dominated river channels, sufficient for sand entrainment but below those required to mobilize gravels (Julien, 2018).

All image-based measurements reported here were collected during the settling phase. Suspended flocs were imaged through a 3 mm-thick flow-through slit built into the tank wall at the observation level. A DSLR fitted with a 5× microscope objective was aligned with the slit to record flocs as they crossed the narrow optical path, enhancing image clarity and increasing the number of focused particles captured by the camera, following methods similar to Osborn et al. (2021).

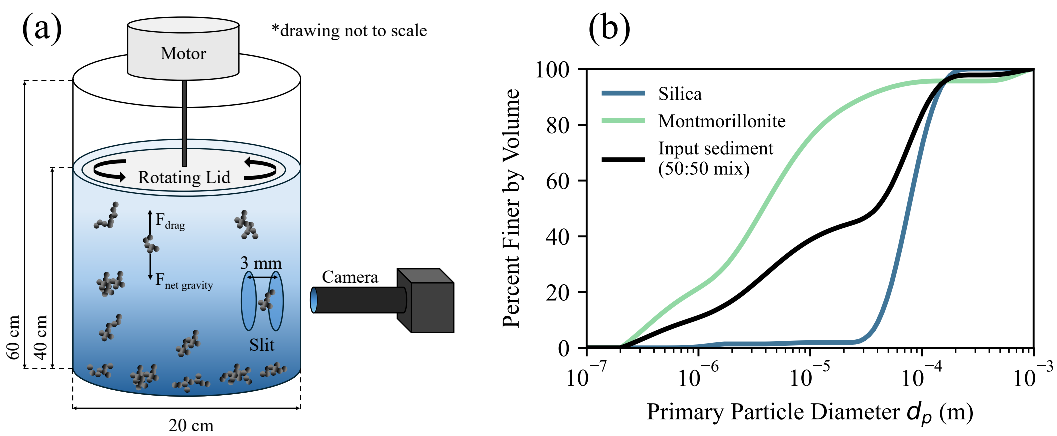

Figure 2. (a) Schematic of the custom flocculation apparatus. A cylindrical PVC tank (20 cm diameter, 60 cm height) is filled to 40 cm depth and stirred with a motor-driven rotating lid just below the surface. An observation slit (3 mm wide) enables in situ imaging of suspended flocs with a DSLR camera and 5× microscope objective aligned for high-resolution focus. (b) Grain-size distributions of the dispersed input sediment measured by the Mastersizer from Nghiem (2025). Curves show pure silica sand (SiO_2), montmorillonite clay, and their 50:50 volume-weighted mixture.

2.2 Experimental Runs

We conducted eight experimental runs in total. The first two were calibration runs using laboratory spheres and silica grains, which do not flocculate, to verify the accuracy of our settling-velocity measurements and provide a baseline for comparing non-flocculating particles to flocs under identical experimental conditions. The remaining six runs systematically varied shear velocity and organic matter (OM) content, given that previous studies have shown that turbulent shear limits floc size by promoting breakup, while organic matter enhances aggregation by binding particles (Nghiem et al., 2022). By independently controlling these variables, we aimed to isolate their respective roles in floc formation and settling.

Each floc experiment used 10 g of solids: 5 g silica (SiO_2) and 5 g montmorillonite clay. Organic content was varied in three runs by adding guar gum, a polysaccharide proxy for plant-derived extracellular polymers, at 1, 2, or 3 wt% relative to the total mass of solids, following the organic-loading framework of Zeichner et al. (2021). This range was selected to (i) remain within the lower end of organic fractions reported for natural river suspended sediment (order 1–6% and up to $\sim$20% in compiled field datasets), and (ii) avoid polymer-dominated regimes while still producing a resolvable change in flocculation dynamics. In Zeichner et al. (2021), polymer-to-clay ratios were chosen explicitly to span modern river organic-matter concentrations, and their experiments show that even 1–2 wt% guar gum produces a strong flocculation response. Guar gum is a controlled proxy for natural extracellular polymeric substances and does not capture the full chemical complexity of field organic matter. Full experimental conditions, including organic-matter fraction and imposed u_*, are listed in Table 1.

Table 1. Summary of experimental conditions.

Exp Num	Experiment Type	OM (wt%)	Shear velocity (m/s)
1	Lab Sphere	NA	NA
2	Silica	NA	NA
3	Floc Shear 1	2	0.02
4	Floc Shear 2	2	0.03
5	Floc Shear 3	2	0.04
6	Floc OM 1	1	0.04
7	Floc OM 2	2	0.04
8	Floc OM 3	3	0.04

NA: Not applicable.

Grain-size distributions for the silica and montmorillonite were measured using a Malvern Mastersizer 3000 (Fig. 2b), following the procedure of Nghiem (2025). In brief, samples were dispersed in deionized water, sonicated to break up aggregates, and analyzed by laser diffraction to obtain particle-size distributions. Together, these two minerals span nearly three orders of magnitude in diameter, with primary particles ranging from 2×10^{-7} to 2×10^{-4} m (percent finer by volume). For the mixed-sediment runs, a composite distribution was calculated as the 50:50 volume-weighted average of the two minerals.

2.3 Image Processing

The floc tracking analysis began with processing every frame of the recorded video sequence, acquired at a constant frame interval $\Delta t = 1/30$ s, as read from the video metadata, so that the sampling frequency equals the acquisition rate. Each frame was converted to an eight-bit grayscale image using OpenCV, and a time-averaged background image (computed by averaging a block of 1000 consecutive frames) was subtracted from each image to isolate transient features such as moving flocs (Fig. 3a).

Particle boundaries in each image were detected from intensity gradients using a global threshold of 200 applied to the normalized residual image. To retain only sharp

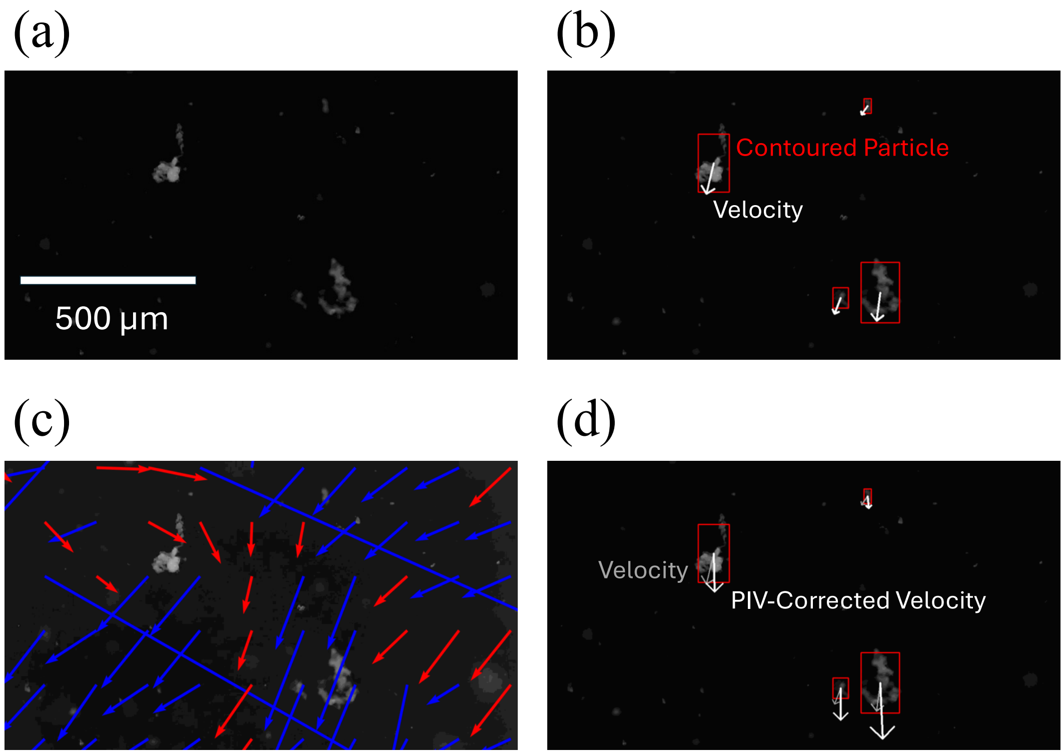

Figure 3. Image processing workflow used for floc analysis. (a) Example of a background-subtracted image, highlighting transient objects such as flocs. (b) Detected and contoured particles passing focus and sharpness thresholds (outlined in red) with raw velocity vectors overlaid. (c) Local background flow field calculated by PIV from tracer particles. (d) Example of a tracked floc with both its raw and corrected (background-subtracted) settling velocity vectors indicated.

and well-focused flocs, candidate objects were required to exceed a Laplacian variance threshold of 5 (measuring focus/sharpness) and to have an intensity standard deviation greater than 5 within their contour (Pertuz et al., 2013). These specific threshold values were optimized for our experimental conditions and image resolution and may vary between different setups. Only particles meeting both criteria were accepted for further analysis and are highlighted in red in Fig. 3b. This objective, per-frame filtering mitigates focus-related sampling bias by excluding blurred detections rather than allowing size-dependent blur to enter the floc statistics. For each accepted object, we recorded the planform area in image pixels A_{px} (later converted to an equivalent diameter after spatial calibration) and the centroid coordinates (x_c, y_c) (for subsequent velocity estimation).

To express diameters and velocities in physical units, we calibrated pixel size using a precision stage micrometer imaged under the same optical conditions as the experiments. Comparing the pixel spacing between micrometer markings with their true separations yielded a scale factor of 0.45 $\mu\mathrm{m\,px^{-1}}$. We applied this factor to all subsequent analyses to convert particle dimensions and inter-frame displacements from pixels to micrometers; velocities were then obtained from these displacements using the recorded frame interval.

Settling velocities were obtained by frame-to-frame tracking using a conservative nearest-neighbor association. For each detection in frame t, we searched within a 50-pixel radius in frame $t{+}1$ for candidate matches. We scored each candidate j using two equally weighted, normalized differences: the equivalent concave-hull diameter d_{ch} (px) and the Laplacian-variance sharpness L (arb. units). The score was defined as $S_j = \frac{1}{2}\big(|d_{\mathrm{ch},j} - d_{\mathrm{ch},t}|/d_{\mathrm{ch},t} + |L_j - L_t|/L_t\big)$. We selected the candidate with the smallest S_j, provided that both relative differences were ≤ 0.10; otherwise, the track was terminated to avoid spurious links from rapid shape or focus changes. The vertical velocity for that interval was initially computed as $\Delta y\, s/\Delta t$, where Δy is the vertical centroid shift (px, positive downward), s is the spatial calibration (0.45 $\mu\mathrm{m\,px^{-1}}$), and Δt is the frame interval (s). Raw velocities computed in $\mu\mathrm{m\,s^{-1}}$ were subsequently converted to $\mathrm{m\,s^{-1}}$ to match the physical units required by Eq. (1) and Eq. (4).

Frame-to-frame particle displacements yield raw 2D velocity vectors that combine gravitational settling with background flow in the imaging plane. To isolate the settling signal, we first computed a local 2D flow field using OpenPIV (Liberzon et al., 2021) for each frame pair. Following Smith and Friedrichs (2015), the images were digitally filtered to retain only tracers smaller than 10 μm that behave as passive flow followers (Fig. 3c). For each tracked floc, we then subtracted the corresponding local flow vector—bilinearly interpolated at the floc centroid—from the floc's raw trajectory vector. This vector differencing produces a background-corrected motion (Fig. 3d, gray arrow for raw, white arrow for corrected), from which we report the vertical component as the settling velocity w_s. All settling velocities in this study refer to this corrected, background-subtracted quantity.

After velocity correction, floc diameters were derived from each particle's projected area. Two common image-based metrics are used to estimate diameter: the Feret diameter (maximum caliper length across the projection) and the equivalent circular diameter (ECD), defined as the diameter of a circle with the same projected area (Jarvis et al., 2005; Smith & Friedrichs, 2015). The Feret diameter is sensitive to orientation and outliers, often overestimating size for irregular or elongated flocs. By contrast, the area-equivalent diameter—especially when computed from a concave hull—can better represent irregular shapes by de-emphasizing extreme protrusions, though it may underrepresent thin filaments. Accordingly, we adopt the concave-hull ECD (with $\alpha = 0.05$, the concave-hull radius parameter that controls the allowed boundary indentation), corresponding to the projected-area–equivalent diameter of Smith and Friedrichs (2015), while acknowledging that alternative definitions are reasonable (Jarvis et al., 2005). Particles were segmented with binary masks, boundaries were traced using a concave-hull algorithm, the enclosed area was measured, and the diameter of a circle of equal area was computed and used as d_f in all subsequent log–log analyses. Because d_f is defined geometrically from the 2D projection, this step does not assume constant floc density; any density changes associated with size or organic content enter only through the measured settling velocities.

2.4 Calibration and Settling Analysis

We calibrated our setup with Experiments 1 and 2, which measured the settling of laboratory spheres (50 µm) and natural silica grains (Fig. 4a). Settling velocities were fitted with Stokes' law (Eq. (1)), treating the shape factor b_1 as a free parameter. For the spheres, the best-fit value $b_1 = 17.7$ is consistent with the theoretical value of 18 for smooth spheres (Fig. 4b), which lies within the confidence interval of the fit. This agreement indicates that the experimental setup accurately reproduces Stokes' settling behavior for smooth spheres. In contrast, silica grains settled more slowly, yielding $b_1 = 29.2$, a drag increase attributable to their angular, non-spherical shapes; Ferguson and Church (2004) reported a comparable $b_1 \approx 24$ for natural sediments of similar morphology.

Compared with silica, flocs settled more slowly and showed substantially greater dispersion in velocity at a given diameter (Fig. 4b). This reduced settling reflects the lower effective density of flocs arising from their highly porous, irregular internal structure (Kranenburg, 1994; Winterwerp, 1998). Consequently, the floc data are not described by the standard Stokes formulation (Eq. (1)) and instead require the floc-specific relation (Eq. (4)), which introduces the 3D fractal dimension n_f and the primary particle diameter d_p as additional parameters.

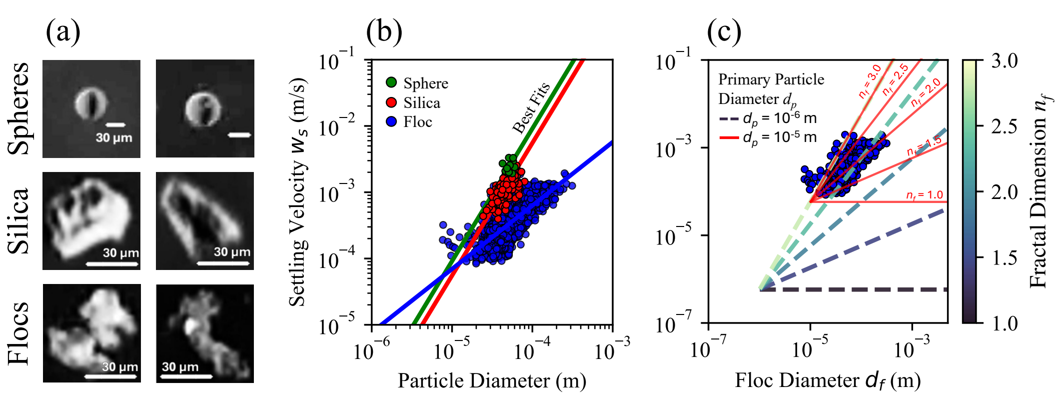

Figure 4. (a) Images of a laboratory sphere (Exp. 1), silica grain (Exp. 2), and a representative floc (Exp. 5). The 30 μm scale bar applies to all. (b) Log–log plot of settling velocity (w_s) vs. diameter (d_f) for spheres (green circles, Exp. 1), silica (red circles, Exp. 2), and aggregated flocs (blue circles, Exps. 3–8). Sphere and silica fits use Stokes' law (Eq. (1)), yielding shape factors (b_1) of 17.7 and 29.2. The floc data fit the fractal settling model ($w_s \propto d_f^{n_f-1}$) with a slope of 0.90, giving an apparent fractal dimension $n_f = 1.90$. (c) Sensitivity of the fractal settling model (Eq. (4)) to assumed primary particle diameter (d_p). The floc data are compared to theoretical predictions using $n_f = 1.9$ for a primary particle size (d_p) of 10^{-6} m (dashed line) and 10^{-5} m (solid red line). This highlights how the choice of d_p controls the required model fit, a key uncertainty in the conventional approach.

However, directly calculating n_f by fitting Eq. (4) to the floc data is problematic because the primary-particle diameter, d_p, is unknown. Although our experiments have known distributions of input particle sizes (Fig. 2b), we do not know which particle sizes are actually incorporated into any given floc. Even if we could measure the primary particle sizes directly, the fractal theory necessitates specifying only a single value for a characteristic primary particle size, and it is unclear how to define this value from a distribution. Moreover, because n_f appears both as an exponent on d_f and within the pre-factor of Eq. (4) (via $d_p^{-(n_f-3)}$), a conventional regression that fits the model to the data for a prescribed d_p (a one-parameter fit) forces the inferred n_f to mathematically compensate for the assumed pre-factor. Consequently, the assumed value for d_p exerts a strong influence on the inferred fractal dimension n_f (Fig. 4c): assuming $d_p = 1 \times 10^{-6}$ m yields $n_f = 2.77$; $d_p = 1 \times 10^{-5}$ m gives $n_f = 2.10$; and $d_p = 1 \times 10^{-4}$ m produces $n_f = 3.00$.

To avoid explicitly solving for the unknown primary-particle size d_p, we use the logarithmic form of Eq. (4): $\log(w_s) = (n_f - 1)\log(d_f) + C$. Here, the intercept C lumps together d_p, the shape factor b_1, and other constants. Under the assumption that d_p and b_1 are approximately constant across the fitted population, the slope in $\log(w_s)$–$\log(d_f)$ space yields $n_f = \text{slope} + 1$ without requiring prior knowledge of d_p. For the full floc population, the fitted slope is 0.90, giving $n_f = 1.90$ (Fig. 4b). However, if structural heterogeneity causes n_f to covary with size, or if effective d_p varies across the population, a bulk regression can exhibit Simpson-type data-aggregation bias. We therefore stratify flocs by their in situ, image-based fractal dimension and fit separate regressions within each stratum; the resulting slopes provide subgroup-specific estimates of n_f and reveal systematic dependence on structure.

2.5 Image-based 2D box-counting dimension

Fractal theory relates the fractal dimension to the primary particle diameter, the number of primary particles, and the floc diameter (Eq. (2)). In natural systems, however, neither d_p nor N is constant or directly measurable during experiments. As a result, numerous methods have been developed to estimate floc fractal dimensions from image-based properties. These approaches are inherently 2D, yielding an apparent fractal dimension that may differ from the hydrodynamically relevant 3D value. Throughout this paper, we denote the 3D fractal dimension inferred from settling velocity scaling as n_f, and use $n_f^{(2D)}$ for the 2D box-counting dimension measured from images. Where the specific 2D method matters (Methodology), we distinguish the box-counting dimension $n_{f,\,\text{box}}^{(2D)}$ from the perimeter–area dimension $n_{f,\,\text{perimeter}}^{(2D)}$; in the Results, all 2D values refer to box-counting and are written simply as $n_f^{(2D)}$.

The perimeter–area method has previously been used to estimate a 2D fractal dimension of flocs (Lee & Kramer, 2004; Maggi & Winterwerp, 2004; Tang & Maggi, 2015). This method relies on the empirical scaling relation:

$$P \propto A^{n_{f,\,\text{perimeter}}^{(2D)}/2} \tag{5}$$

where P is the measured perimeter and A is the projected area of the particle in the 2D image. In practice, $n_{f,\,\text{perimeter}}^{(2D)}$ is estimated from the log–log slope of P versus A. This expression is derived by analogy to the Euclidean relation $P = kA^{1/2}$, which holds for smooth (non-fractal) boundaries with $n_{f,\,\text{perimeter}}^{(2D)} = 1$. The underlying assumption is that, for a fractal boundary, the exponent in this scaling law approximates the true boundary fractal dimension $n_{f,\,\text{perimeter}}^{(2D)}$. However, both mathematical analysis and geometric constructions (e.g., the Koch curve) demonstrate that this assumption breaks down for truly fractal sets (Cheng, 1995; Frame et al., 2025). For such boundaries, the measured perimeter is strongly dependent on the image resolution, making a simple P–A scaling ill-defined without explicitly accounting for the measurement scale. As a result, perimeter–area scaling does not reliably recover the true fractal dimension from 2D projections.

Box-counting, in contrast, is an appropriate method for fractal-dimension estimation that does not rely on strict self-similarity and is computationally straightforward (Foroutan-pour et al., 1999). A key advantage of box-counting is that it does not require knowledge of the primary particle size (d_p) or the number of constituent particles (N), making it particularly well-suited for natural flocs where these values are unknown or variable (Bellouti et al., 1997; Spencer et al., 2022). The box-counting dimension $n_{f,\,\mathrm{box}}^{(2\mathrm{D})}$ is determined by measuring how the number of occupied boxes $N_{\mathrm{box}}(\epsilon)$ at a given scale ϵ varies with scale (Mandelbrot, 1967):

$$N_{\mathrm{box}}(\epsilon) \propto \epsilon^{-n_{f,\,\mathrm{box}}^{(2\mathrm{D})}} \tag{6}$$

For each floc mask, we computed $N_{\mathrm{box}}(\epsilon)$ using square boxes with side lengths ϵ spanning powers of two in pixel units ($\epsilon = 1, 2, 4, \ldots$), and we estimated the dimension from the slope of a least-squares fit over the portion of the log–log curve that remains approximately linear and spans at least one order of magnitude in scale. When analyzing images of objects with finite extent from in situ imaging, finite-size effects and image pixelation make the estimated fractal dimension sensitive to the available scaling range. For a digital image of $W \times W$ pixels, the practical range of box sizes (ϵ) spans from 1 to W pixels. Reliable estimation of $n_{f,\,\mathrm{box}}^{(2\mathrm{D})}$ requires a sufficiently broad range of box sizes to establish a stable linear trend on a log–log plot of $N_{\mathrm{box}}(\epsilon)$ versus $1/\epsilon$. If the image is too small, the limited scaling range can yield unstable values of $n_{f,\,\mathrm{box}}^{(2\mathrm{D})}$. For example, a 20×20 pixel image provides only about 1.3 orders of magnitude in scale, which is insufficient for robust fitting and produces scale-dependent estimates (Fig. 5a). In contrast, a 100×100 pixel image spans two full orders of magnitude and typically provides 6–7 usable data points for regression; beyond this image size, the estimated $n_{f,\,\mathrm{box}}^{(2\mathrm{D})}$ saturates, indicating a practical threshold for resolution-independent estimation (Fig. 5a). Accordingly, all analyzed flocs were extracted using 100×100 pixel windows centered on each particle. Fig. 5b shows examples for two experimental flocs at high image resolution, illustrating the application of the box-counting method to real data. In this study, all reported values of the 2D box-counting dimension ($n_{f,\,\mathrm{box}}^{(2\mathrm{D})}$) in the results and discussion sections refer exclusively to measurements obtained by the box-counting method.

2.6 Primary Particle Diameter (d_p)

In most in situ experiments, the primary particle diameter d_p is assumed from the input sediment-size distribution and hypothesized floc-formation conditions because d_p is rarely resolvable directly. This practice can be uncertain when the effective primary scale differs from the feed distribution or varies across aggregation pathways. Here we identify d_p empirically by leveraging multiple floc subpopulations with distinct fractal dimensions n_f. In the experimental application, we treat d_p as an experiment-level parameter common to all n_f-stratified subpopulations. As implied by Eq. (4), d_p does not affect the slope of the $\log(w_s)$–$\log(d_f)$ relation (the slope is n_f-1), but it fixes a common intersection of the subgroup regressions. This is because at $d_f = d_p$, the object is a primary particle (not a floc), so substituting $d_f = d_p$ into Eq. (4) yields $w_s = (g\,R_s/b_1\nu)\,d_p^2$, which is independent of n_f. Consequently, regressions from any number of n_f-stratified subpopulations intersect at $(d_f, w_s) = (d_p, (g\,R_s/b_1\nu)\,d_p^2)$. We refer to this common intersection diameter as $d_{\mathrm{intersect}}$. In our synthetic example (Fig. 1), three subgroups with $n_f \approx 1.5$, 2.0, and 2.5 produce regression lines that converge at $d_{\mathrm{intersect}} \approx 10^{-5}$ m, because the synthetic data were generated using this prescribed primary-particle diameter across varying fractal dimensions, thereby illustrating the method.

A floc contains many primary particles. For example, with $d_f \sim 10^{-4}$ m, $d_p \sim 10^{-5}$ m, and $n_f \approx 2$, Eq. (2) gives $N \sim (d_f/d_p)^{n_f} \sim 10^2$–$10^4$. This means the set-

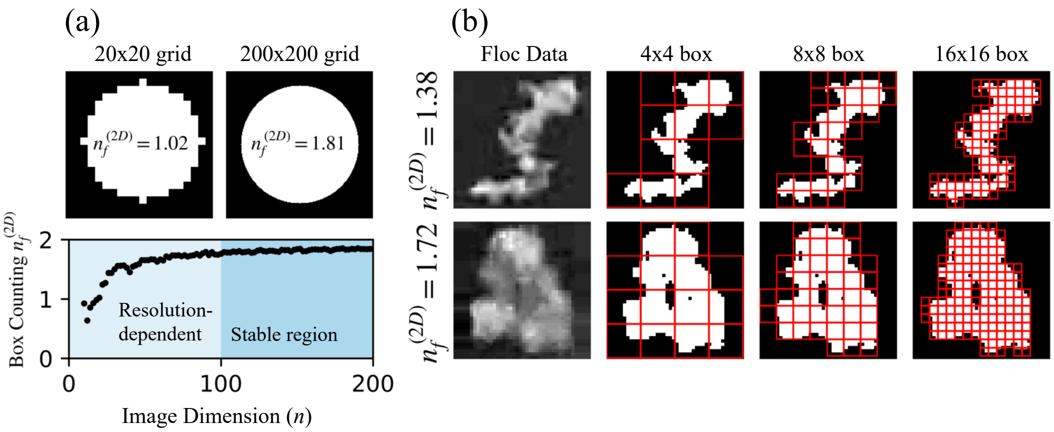

Figure 5. (a) Illustration of resolution dependence in box-counting dimension estimation. Top: A perfect circle embedded in 20 $\times$ 20 and 200 $\times$ 200 pixel grids shows that coarse grids systematically underestimate the fractal dimension. Bottom: Plot of measured 2D box-counting dimension $n_{f,\,\mathrm{box}}^{(2\mathrm{D})}$ versus image width W, highlighting a resolution-dependent regime at low W and a stable region at high W. The transition scale ($W \approx 100$ pixels) was identified by repeating the box-counting calculation across increasing image sizes and selecting the smallest W beyond which further increases produced negligible changes in the estimated dimension. (b) Example box-counting dimension estimates for two experimental flocs imaged at $> 100 \times 100$ pixel resolution.

tling measurements reflect the collective behavior of aggregates built from a common sediment pool, and the inferred d_p should be interpreted as an effective primary scale for the flocs resolved by the imaging and tracking. In the synthetic example, the subgroup regressions intersect exactly at a single point because the data are generated under the idealized assumption that all primary particles share a single diameter d_p. In natural sediments, primary particles span a distribution of sizes, so different flocs can be built from differently sized particles and the subgroup regressions need not converge to a unique intersection. We therefore treat the intersection as distributed rather than singular and use a Monte Carlo procedure to generate an ensemble of intersection points, which we interpret as a distribution of effective d_p values. We then compare this inferred d_p distribution to the input sediment-size distribution to assess consistency and to quantify how the sediment's primary-size variability propagates into the settling-based estimate of d_p.

The intersection method assumes a common shape factor b_1 across fractal subgroups. Because drag depends on aggregate structure, b_1 may vary with n_f, biasing the inferred intersection diameter. Allowing for subgroup-dependent drag, the relationship between the observed intersection diameter $d_{\mathrm{intersect}}$ and the true primary particle diameter d_p for two subpopulations j and k is

$$d_{\mathrm{intersect}} = d_p \left(\frac{b_{1,j}}{b_{1,k}} \right)^{1/(n_{f,j} - n_{f,k})}, \tag{7}$$

which reduces to

$$d_{\mathrm{intersect}} = d_p \tag{8}$$

when $b_{1,j} = b_{1,k}$. In this limit, all subgroup regressions intersect exactly at the primary particle diameter.

3 Results

3.1 Floc Bulk Properties

We first report the bulk distributions of floc diameter (d_f) and settling velocity (w_s) across all experimental conditions (Fig. 6). Median d_f decreases monotonically with increasing shear velocity (Fig. 6a), consistent with stronger turbulence breaking apart aggregates. In contrast, d_f responds weakly and non-monotonically to organic matter content, increasing slightly from 1% to 2% OM before declining at 3% OM (Fig. 6b). Median settling velocity w_s exhibits a non-monotonic dependence on shear velocity, peaking at $u_* = 0.03\ \mathrm{m\,s^{-1}}$ rather than decreasing steadily (Fig. 6c). With increasing organic matter, w_s decreases monotonically (Fig. 6d).

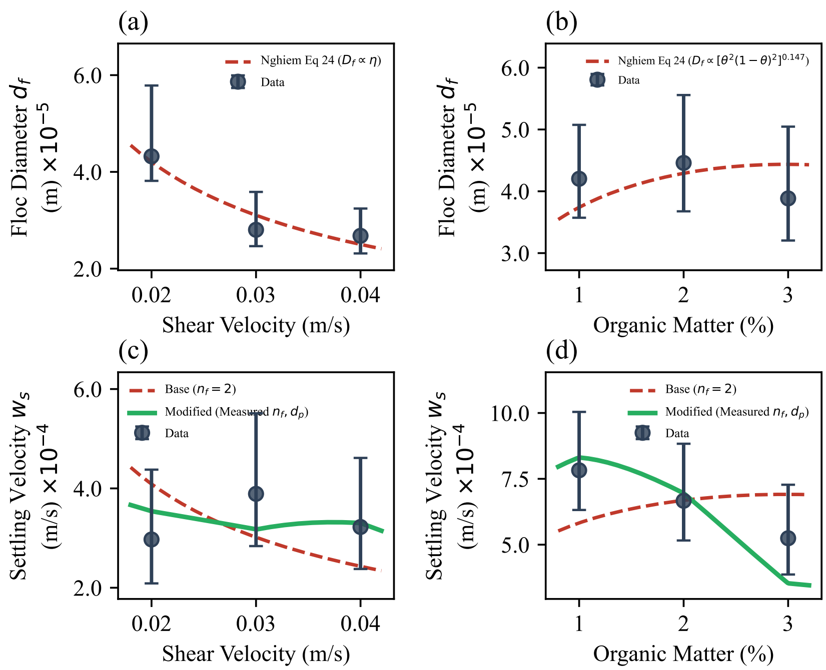

Figure 6. Median floc diameter (d_f) and settling velocity (w_s) with interquartile ranges for varying shear velocity (a, c) and organic matter content (b, d). Dashed lines show normalized model predictions: in panels (a, b), the Nghiem et al. (2022) equilibrium floc diameter trend $d_{f,\mathrm{eq}} = f(\eta, \theta)$, which is independent of n_f; in panels (c, d), the base model with $n_f = 2$ (red dashed; Eq. (4) with $n_f = 2$) and the modified model with measured n_f and d_p (blue dashed; Eq. (4) with condition-specific n_f and d_p; see Section 3.4). Each model trend is normalized by a single least-squares constant.

To interpret these trends mechanistically, we compare our observations against the semi-empirical freshwater flocculation framework of Nghiem et al. (2022). That model predicts both the equilibrium floc diameter d_f and the corresponding settling velocity w_s as functions of two environmental inputs: the Kolmogorov microscale η (which sets the smallest turbulent eddy that can break flocs) and the fractional organic surface coverage θ (which controls binding strength). Evaluating this model against our data requires mapping our experimental control variables—imposed shear velocity u_* and organic matter weight fraction—to η and θ.

In applying their model to river field data, Nghiem et al. (2022) used an open-channel relation where turbulent dissipation is generated solely by bed friction. Our experimental apparatus is a top-driven cylindrical tank in which stationary sidewall friction acts as an additional dominant energy sink. To calculate the volume-averaged turbulent dissipation rate $\bar{\varepsilon}_{\text{tank}}$, we evaluate the steady-state power balance (Camp & Stein, 1943), equating the power injected by the rotating lid to the power dissipated by friction on all tank boundaries (bed and sidewalls). Assuming the wall shear stress scales with the calibrated bed shear velocity ($\tau_w \approx \rho_w u_*^2$) (Pope, 2000; Schlichting & Gersten, 2000) and substituting the bulk slip velocity $U_{\text{bulk}} \approx u_* \sqrt{2/C_f}$, the mean dissipation rate is:

$$\bar{\varepsilon}_{\text{tank}} = \frac{u_*^3}{\sqrt{C_f/2}} \left(\frac{1}{h} + \frac{2}{R} \right), \tag{9}$$

where $h = 0.40$ m is the water depth, $R = 0.10$ m is the tank radius, and $C_f \approx 0.005$ is the skin friction coefficient for fully turbulent swirling flow (Daily & Nece, 1960; Schlichting & Gersten, 2000). The Kolmogorov microscale is then $\eta = (\nu^3/\bar{\varepsilon}_{\text{tank}})^{1/4}$. Because of the extensive sidewall area, $\bar{\varepsilon}_{\text{tank}}$ is substantially larger than open-channel predictions for an equivalent u_*, so the absolute value of η is smaller. However, $\bar{\varepsilon}_{\text{tank}} \propto u_*^3$ regardless of tank geometry, so the functional scaling is identical to an open channel: $\eta \propto u_*^{-3/4}$.

Nghiem et al. (2022) parameterized organic binding using field measurements of particulate organic carbon (OC), requiring an intermediate stoichiometric assumption (cellulose composition) to convert OC to total organic matter (OM). Because our experiments use guar gum—a pure polysaccharide proxy for natural extracellular polymeric substances—we bypass this conversion and use the experimental weight percentage of OM directly. Following Eq. 19 of Nghiem et al. (2022), the fractional surface coverage is:

$$\theta = \frac{(\%\text{OM}/100)\,(\rho_s/\rho_{\text{OM}})\,d_p^3}{(d_p + \delta)^3 - d_p^3}, \tag{10}$$

where $\rho_s = 2650\ \text{kg}\,\text{m}^{-3}$ is the sediment grain density, $\rho_{\text{OM}} = 1000\ \text{kg}\,\text{m}^{-3}$ is the organic matter density, d_p is the primary particle diameter, and $\delta = 1\ \mu$m is the assumed organic coating thickness.

A key property of the Nghiem et al. (2022) framework is that the predicted equilibrium floc diameter does not depend on the fractal dimension. In their model (Eq. 9 in their study), the fractal dimension n_f enters both the aggregation and breakup terms of the floc growth equation through identical powers of d_f: aggregation scales as $d_f^{n_f+2}$ and breakup as $d_f^{n_f}$. When these two terms are set equal at equilibrium, n_f cancels algebraically and the resulting equilibrium diameter $d_{f,\text{eq}}$ depends only on η and θ. Physically, this cancellation reflects the fact that the same fractal structure governs both how efficiently particles collide (aggregation) and how susceptible the aggregate is to turbulent stresses (breakup), so the two effects offset exactly at equilibrium.

Using the calibrated exponents from Nghiem et al. (2022) (their Eqs. 23–24), the equilibrium diameter scales as $d_{f,\text{eq}} \propto \eta^r[\theta^2(1-\theta)^2]^q$, where r and q are fitted constants. Since $\eta \propto u_*^{-3/4}$, the predicted shear dependence is $d_{f,\text{eq}} \propto u_*^{-0.75r}$, and the organic-matter dependence enters through θ (Eq. (10)). Because the model parameters were calibrated for river field conditions rather than our tank geometry, we compare trends rather than absolute magnitudes: we normalize the predicted $d_{f,\text{eq}}$ by a single least-squares constant and plot it against the measured medians in Fig. 6a,b; the model captures the monotonic decrease of d_f with shear and the weak non-monotonic response to organic matter.

Although the equilibrium floc diameter is independent of n_f, the settling velocity is not. Equation (4) shows that w_s depends on n_f through two exponents: $d_p^{3-n_f}$ in the prefactor and $d_f^{n_f-1}$ in the size scaling. The Nghiem et al. (2022) model adopts $n_f =$

2 throughout, which simplifies Eq. (4) to $w_s \propto d_p\, d_f$, a linear dependence on floc diameter in which d_p and b_1 are absorbed into the prefactor. We refer to this as the base model. We compare the predicted and observed trends rather than absolute magnitudes for two reasons. First, the Nghiem et al. (2022) model parameters were calibrated against river field data; applying them to our tank geometry, which has a different energy dissipation distribution, would require re-calibration. Second, the settling velocity prefactor contains the shape factor b_1, which is not independently constrained and may itself vary with n_f (see Section 3.3). We therefore normalize each model prediction by a single least-squares constant ($\hat{w}_{s,\mathrm{base}} = k_{\mathrm{base}}\, d_{f,\mathrm{eq}}$) and compare the predicted trend against the measured medians in Fig. 6c,d. Because the base model is linear in d_f, and d_f decreases monotonically with shear, the base model strictly predicts a monotonic decrease in w_s with increasing u_*. It therefore cannot reproduce the non-monotonic peak observed at $u_* = 0.03\ \mathrm{m\,s^{-1}}$ (Fig. 6c). Similarly, because d_f increases slightly with OM (for $\theta < 0.5$), the base model predicts w_s increasing with OM, opposite to the observed monotonic decline (Fig. 6d).

These discrepancies motivate a closer examination of the internal structural properties—the fractal dimension n_f and the effective primary particle diameter d_p—that the base model holds fixed. In the following two subsections, we develop methods to extract condition-specific estimates of n_f (Section 3.2) and d_p (Section 3.3), and in Section 3.4 we show that incorporating this measured structural variability into the settling velocity prediction resolves the failures identified above.

3.2 Stratifying 2D Box-Counting Data

To infer a fractal dimension from settling data, we group flocs into bins based on their 2D box-counting dimension, $n_f^{(2D)}$, measured from images. The bin width in $n_f^{(2D)}$ serves only as a coarse-graining scale for separating discrete structural classes, not as a physical control parameter. If a population contains distinct structural classes (e.g., non-flocculating grains with $n_f \approx 3$ and flocs with $n_f \approx 2$), finer stratification merely subdivides these classes into redundant bins that recover the same class-specific slopes.

As a sensitivity analysis, we computed the 2D box-counting dimension, $n_f^{(2D)}$, for Experiment 3 (2 wt% organic matter; shear velocity $u_* = 0.02\ \mathrm{m\,s^{-1}}$). To evaluate the impact of stratification, we divided the dataset into bins of width 0.1, 0.05, and 0.025, corresponding to 3, 6, and 12 bins of $n_f^{(2D)}$, respectively, within a manually defined range from 1.3 to 1.6. Within each bin, we fitted a power-law relation between settling velocity, w_s, and floc diameter, d_f, in log–log space (Figs. 7a, d, g). The model $\log_{10}(w_s) = m\log_{10}(d_f) + C$ was calibrated using an ensemble Markov chain Monte Carlo (MCMC) sampler to obtain robust parameter estimates and verify convergence. After burn-in and thinning, the posterior medians of the slope m were converted to the hydrodynamically relevant 3D fractal dimension, denoted simply as n_f, using the relation $n_f = m + 1$.

Plotting the inferred 3D fractal dimension, n_f, against the corresponding group-mean 2D value, $n_f^{(2D)}$, shows how the stratification behaves across binning schemes. A linear trend captures the binned data reasonably well for all cases (Figs. 7c, f, i). However, because the number of bins is small (3, 6, or 12), the resulting coefficients of determination are not themselves a robust metric of model adequacy, as high R^2 values are expected even in the presence of substantial uncertainty.

A more informative assessment comes from quantifying the dynamical consequences of refining the stratification. From Eq. (4), assuming the log–log regression lines converge near the primary particle diameter d_p, a change Δn_f produces a vertical shift $\Delta\log_{10}(w_s) \approx \Delta n_f \log_{10}(d_f/d_p)$. Over the observed range of macroscopic floc sizes ($d_f/d_p \approx 10^1$–10^2), a change of $\Delta n_f = 0.01$ produces a shift of $\Delta\log_{10}(w_s) \approx 0.01$–$0.02$, corresponding to a change in settling velocity of less than 5%. This effect is minimal relative to measurement scatter. Consequently, increasingly fine stratification does not reveal qualitatively different settling regimes but instead primarily inflates variance in the estimated slopes as bin populations decrease. We therefore adopt a bin width of 0.1 for subsequent analyses. This choice resolves meaningful structural heterogeneity while maintaining sufficient sample sizes for regression, and avoids over-resolving differences that have negligible impact on settling velocities over the observed floc-size range.

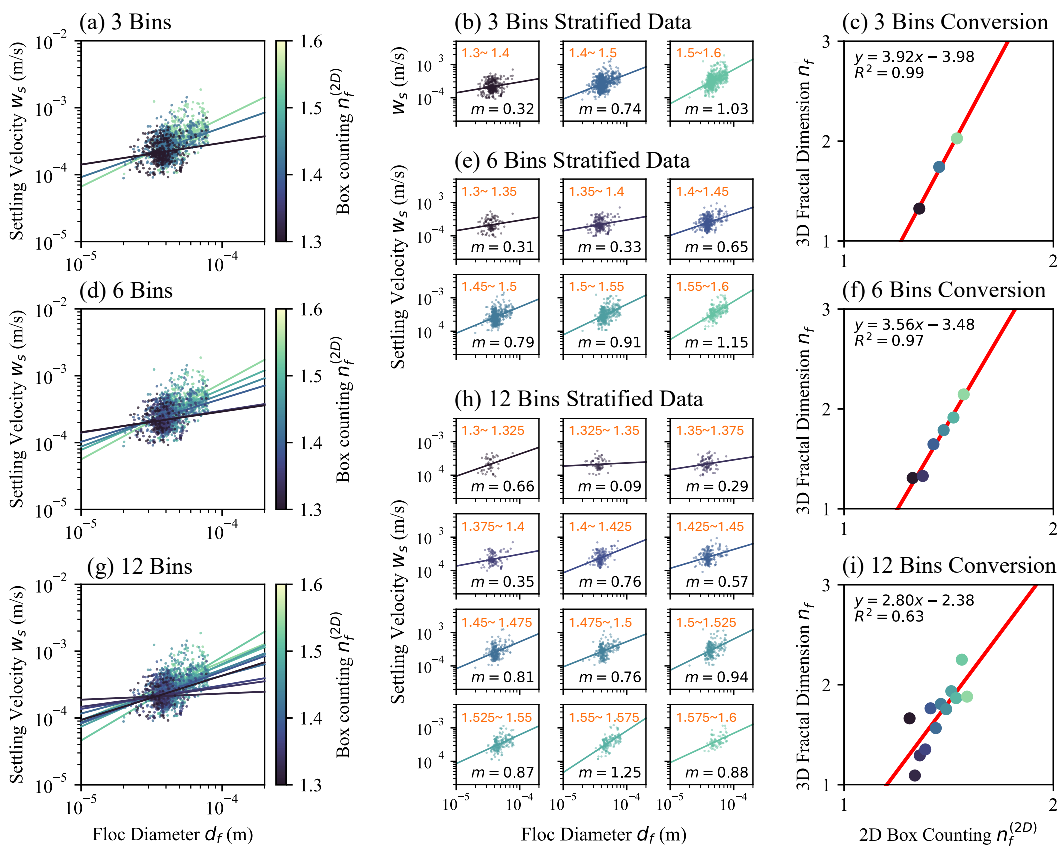

Figure 7. Stratification of floc data by 2D box-counting dimension ($n_f^{(2D)}$) for Experiment 3, illustrating how binning strategy influences the estimation of 3D fractal dimensions. (a–c) Analysis using 3 bins: (a) log–log plots of settling velocity (w_s) versus floc diameter (d_f), color-coded by 2D box-counting dimension bin; (b) fitted slopes (m) for each bin; (c) resulting 3D fractal dimension ($n_f = m + 1$) plotted against mean $n_f^{(2D)}$ for each bin, showing a strong linear relationship. (d–f) Same analysis using 6 bins; (g–i) using 12 bins, highlighting increased scatter and reduced regression stability with finer binning.

The six floc experiments (Experiments 3–8) comprise approximately 12,500 individual floc particles. Within each 0.1-wide $n_f^{(2D)}$ bin, outliers were removed using the interquartile range criterion $[Q_1 - 1.5\,\mathrm{IQR}, Q_3 + 1.5\,\mathrm{IQR}]$, yielding five to eight bins per experiment with a median population of 190 flocs per bin. The same MCMC regression procedure described above was then applied within each bin to obtain n_f.

Figure 8 shows the log–log relationship between settling velocity w_s and floc diameter d_f for experiments with increasing shear velocity (Figs. 8a–c) and increasing organic matter content (Figs. 8d–f). Within each panel, power-law regressions fitted to individual $n_f^{(2D)}$ bins reveal distinct slopes that vary systematically with floc structure, and the corresponding 3D fractal dimensions n_f are obtained via the relation $n_f = m+1$. As the $n_f^{(2D)}$ bin midpoint increases, the recovered n_f also increases monotonically, demonstrating that the 2D structural signature can be used to resolve distinct 3D fractal dimensions within a mixed population. The inset panels illustrate this trend, plotting the bin-median n_f against the $n_f^{(2D)}$ bin midpoint with 95% credible intervals derived from the MCMC posteriors. For comparison, the aggregated (unstratified) regression and its 95% credible interval are shown as the orange dashed line and shaded band, respectively. Notably, the aggregated credible interval does not capture the range of n_f values captured by the stratified bins, indicating that a single power-law regression applied to the bulk population cannot capture the structural heterogeneity present within each experiment.

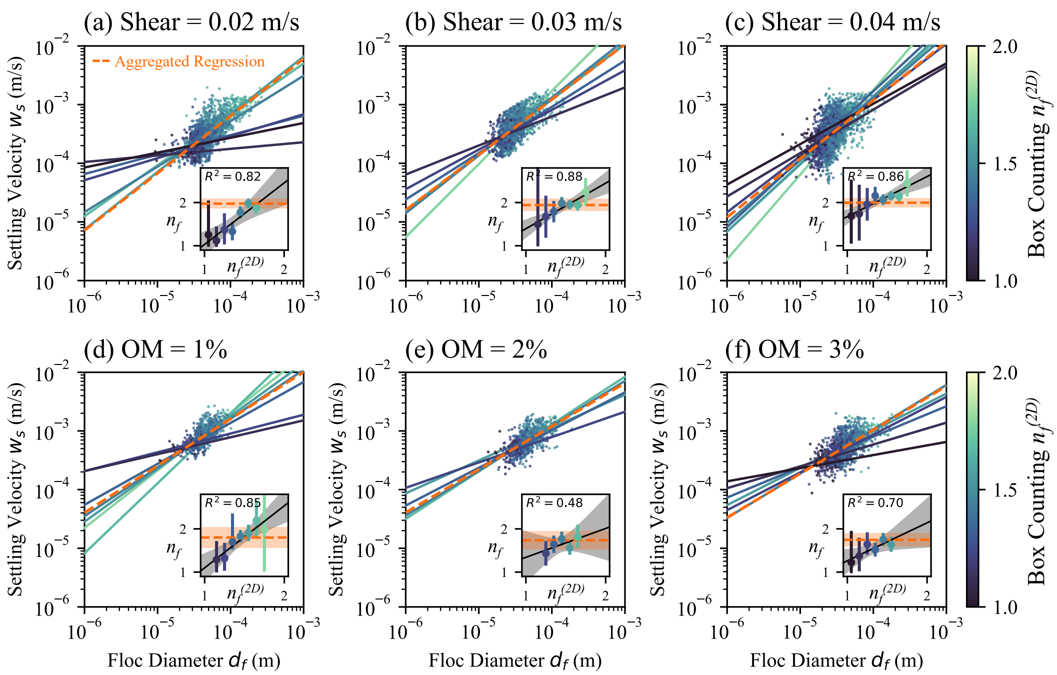

Figure 8. Comparison of stratified and aggregated (bulk) fits for all floc datasets. (a–c) Varying shear velocities of 0.02, 0.03, and 0.04 $\mathrm{m\,s^{-1}}$. (d–f) Varying organic matter contents of 1%, 2%, and 3%. Inset plots show the relationship between $n_f^{(2D)}$ and slope-inferred n_f for each experiment; points represent bin medians (with 95% credible intervals) and are color-coded by $n_f^{(2D)}$ group to make within-population variability visually explicit.

The dependence of the stratified and aggregated n_f estimates on shear velocity and organic matter is summarized in Fig. 9a,b. In the stratified analysis, the interquartile range (IQR) is computed directly from the population of inferred n_f values across bins, weighted by the number of particles contributing to each bin, because each bin represents an independent subset of flocs with distinct fractal structure. In contrast, in the aggregated analysis the IQR is derived from the Bayesian posterior distribution of the fitted slope parameter, since all particles are pooled into a single regression and variability arises from parameter uncertainty rather than particle-to-particle heterogeneity.

Stratified median n_f increases approximately linearly with shear velocity at a rate of $\Delta n_f/\Delta u_* \approx 17\,\mathrm{s\,m^{-1}}$ ($n_f = 17.4\,u_* + 1.42$, Fig. 9a), consistent with shear-driven breakup preferentially destroying loose, low-n_f flocs and favoring compact, high-n_f flocs. The aggregated analysis yields a nearly flat response ($n_f = 1.0\,u_* + 1.94$), demonstrating that the shear-driven structural trend is effectively invisible in bulk data. For organic matter (Fig. 9b), stratified median n_f decreases at $\Delta n_f/\Delta\mathrm{OM} \approx -0.11$ per wt% ($n_f = -0.11\,\mathrm{OM} + 1.92$), indicating that organic binding promotes open, porous structures. The aggregated trend is again much weaker ($n_f = -0.03\,\mathrm{OM} + 1.81$). This insensitivity arises because the aggregated regression fits a single power law through a heterogeneous mixture of floc structures, each with a different w_s–d_f slope. The resulting best-fit slope is dominated by the most populated $n_f^{(2D)}$ bins and by the large scatter inherent in pooling structurally distinct subpopulations, which masks the systematic variation that the stratified analysis resolves. In effect, the noise introduced by collapsing all fractal classes into one regression absorbs the signal that distinguishes compact from porous flocs.

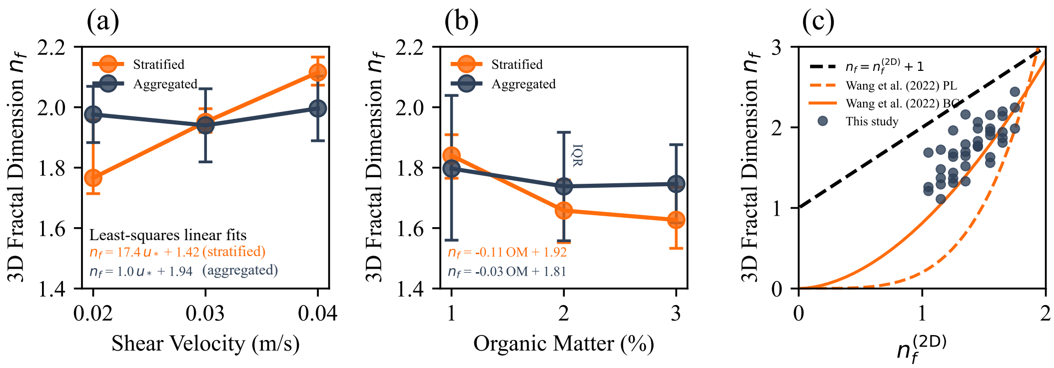

Figure 9. Comparison of stratified and aggregated (bulk) analyses of floc fractal dimensions. (a) Median inferred n_f versus shear velocity. (b) Median n_f versus organic matter (OM) content. Error bars denote interquartile ranges (IQRs) and markers indicate medians; for stratified estimates (orange) the IQR reflects particle-scale variability across bins weighted by population, whereas for aggregated estimates (navy) it reflects uncertainty in the Bayesian posterior of a single fitted parameter. (c) Mapping between 2D and 3D fractal dimensions: black dashed line, theoretical relation $n_f = n_f^{(2D)} + 1$; orange solid line, empirical 2D box-counting to 3D box-counting (BC) conversion ($n_f = 0.81\,(n_f^{(2D)})^{1.81}$); orange dashed line, empirical 2D box-counting to 3D power-law (PL) conversion ($n_f = 0.20\,(n_f^{(2D)})^{4.08}$) (Wang et al., 2022). Experimental data (grey markers) align most closely with the empirical box-counting model.

Figure 9c collects the per-bin data from all six experiments to examine the mapping between $n_f^{(2D)}$ and n_f. The 40 bin-median values (grey circles) are compared against three reference models. The idealized Euclidean relation $n_f = n_f^{(2D)}+1$ (black dashed) assumes perfect space-filling and systematically overpredicts n_f across the measured range. The empirical power-law (PL) and box-counting (BC) conversions of Wang et al. (2022), calibrated on computationally generated aggregates, bracket the data more closely. The BC model ($n_f = 0.8118\,{n_f^{(2D)}}^{1.8054}$, solid orange) tracks the upper envelope of the observations, while the PL model ($n_f = 0.2015\,{n_f^{(2D)}}^{4.079}$, dashed orange) underpredicts at moderate $n_f^{(2D)}$. Our data scatter between these two empirical curves, with the closest overall agreement to the BC conversion. This alignment is consistent with the fact that both our stratification variable and the Wang et al. (2022) BC model are based on box-counting dimensions. Mismatch may arise because the Wang et al. (2022) model constructs flocs from monodisperse primary particles of uniform size and shape, whereas natural flocs comprise polydisperse, irregularly shaped constituents whose heterogeneous packing alters the 2D projection geometry.

3.3 Inferring Primary Particle Diameter (d_p)

The stratified regressions also provide a means of estimating the effective primary particle diameter, d_p. In log–log space, Eq. (4) predicts that regression lines for different n_f converge as $d_f \to d_p$: at the primary-particle scale, a floc reduces to a single grain whose settling velocity is independent of fractal structure. We exploit this convergence by computing all pairwise intersections of the per-bin regression lines from their MCMC posteriors and assume a constant shape factor b_1. Repeating the calculation across thousands of posterior draws yields an ensemble of intersection diameters whose spread reflects both measurement scatter and finite-sample uncertainty. Fig. 10a–c compare the resulting d_p posterior (orange) with the known input grain-size distribution (gray) for the three shear-velocity experiments, and Fig. 10e–g show the corresponding comparison for the three organic-matter experiments.

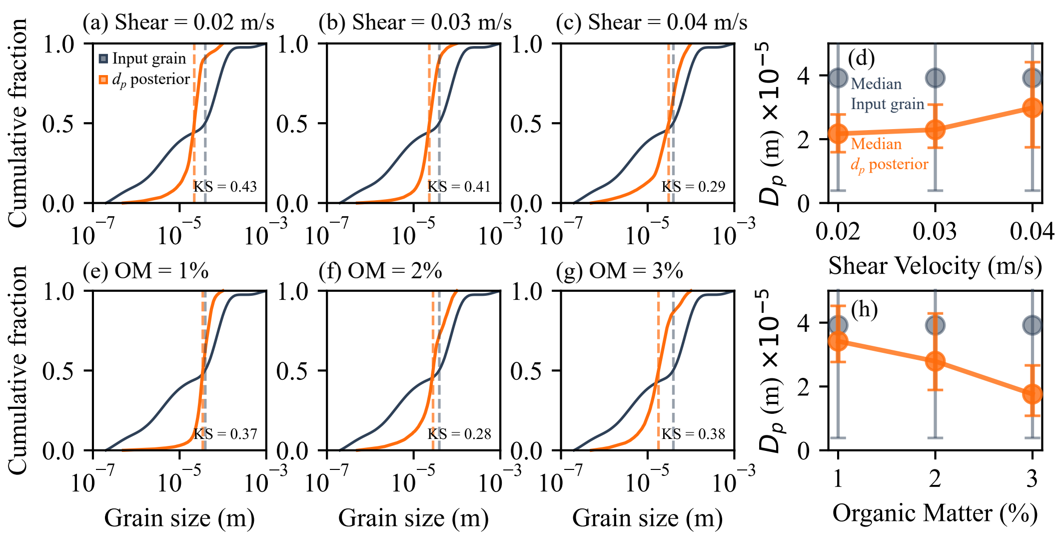

Figure 10. Inferred primary particle diameter (d_p) distributions and summary statistics. (a–c) Cumulative percent finer by volume comparing the known input grain-size distribution (grey) with the inferred d_p posterior (orange) for each shear-velocity experiment; KS statistics quantify agreement. (d) Median inferred d_p (orange, with IQR error bars) versus shear velocity, alongside the input-sediment median and IQR (grey). (e–g) Same as (a–c) for the three organic-matter experiments. (h) Median inferred d_p versus organic matter content.

Results show agreement between the inferred d_p distribution from flocs and the input grain-size distribution, particularly in their central tendency (e.g., alignment of medians) (Fig. 10). The Kolmogorov–Smirnov (KS) statistic ranges from 0.28 to 0.43 across all six experiments, indicating moderate agreement (a KS value of 0 denotes perfect agreement, whereas a value near 1 indicates complete mismatch). Perfect agreement is not expected because the input grain-size distribution in Fig. 10 includes all particle sizes added to the experiments, but not all of these necessarily serve as primary particles in flocs. In every case the inferred d_p distribution is substantially narrower than the input grain distribution, with posteriors peaking in the coarse-silt range (∼10–40 µm), suggesting that the effective building blocks of the observed flocs are drawn predominantly from the silt fraction (Nghiem et al., 2026).

Panels (d) and (h) of Fig. 10 summarize the median d_p (orange, with IQR error bars) alongside the input-sediment median and IQR (grey) across shear velocity and organic matter. With increasing shear velocity (panel d), the inferred d_p increases from

21.7 µm at $u_* = 0.02\ \mathrm{m\,s^{-1}}$ to 23.0 µm at $0.03\ \mathrm{m\,s^{-1}}$ and 29.8 µm at $0.04\ \mathrm{m\,s^{-1}}$, consistent with stronger shear selectively destroying the most fragile aggregates and shifting the effective primary scale toward coarser grains. With increasing organic matter (panel h), the median d_p decreases monotonically from 34.2 µm at 1% OM to 27.9 µm at 2% and 17.6 µm at 3%, indicating that organic binding enables finer particles to participate as effective building blocks. Across all conditions the inferred d_p remains well below the input-sediment median of ∼39 µm (grey markers in panels d and h).

The inferred d_p thus represents an effective primary scale associated with the population of flocs contributing to the measured settling velocities, rather than the full input grain-size distribution. Differences between the input grain-size and inferred d_p distributions reflect both uncertainties in the intersection method and real selective incorporation of certain size classes into flocs. Prolonged turbidity following the experiments indicates that a fraction of very fine material remained in suspension and was not flocculated; however, this observation alone does not distinguish between incomplete aggregation of the smallest particles and limitations of the imaging and tracking workflow at small sizes. Apparent offsets may also arise from size-dependent participation in the floc population resolved by the measurements and from under-sampling of dispersed particles below the detection threshold.

The absolute magnitude of the inferred d_p is sensitive to the assumed value of the shape factor b_1. Laboratory and field studies indicate that b_1 is not constant and may increase with n_f (Strom & Keyvani, 2011). If b_1 varies from $\mathcal{O}(10)$ to $\mathcal{O}(10^2)$ across fractal classes, the inferred intersection diameter $d_{\mathrm{intersect}}$ can differ from the true d_p by up to an order of magnitude for fixed n_f (Eq. (7)). However, because the same $b_1(n_f)$ relationship applies uniformly across all experiments, the relative trends in d_p with shear velocity and organic matter reported above are preserved regardless of the assumed shape-factor model; only the absolute scale shifts.

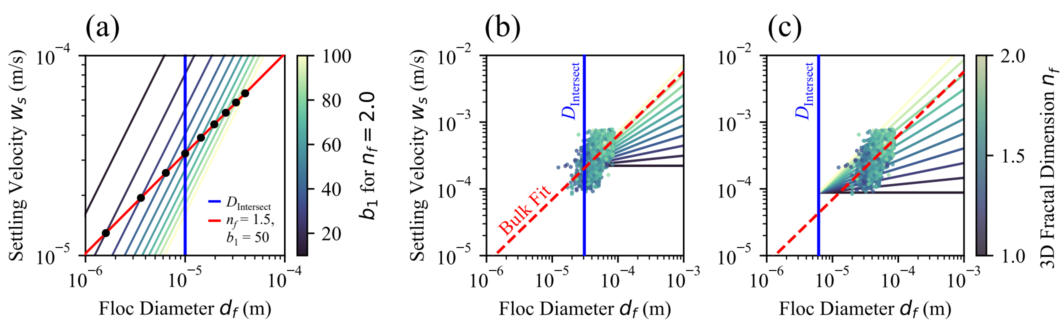

Figure 11. Influence of the shape factor (b_1) on the geometric intersection diameter ($d_{\mathrm{intersect}}$) between floc groups of different fractal dimension. (a) Increasing b_1 with n_f shifts $d_{\mathrm{intersect}}$ above the primary particle diameter d_p, while decreasing b_1 shifts it below. (b) When evaluating the idealized intersection assuming constant b_1, the lower tail of measured floc diameters extends to the left of $d_{\mathrm{intersect}}$. This violates the physical constraint that flocs cannot be smaller than their fundamental building blocks. (c) Example where $d_{\mathrm{intersect}} > d_p$, emphasizing that b_1 must rise with n_f to preserve the proper size hierarchy as denser flocs experience greater drag.

To illustrate this effect, we plot settling velocity versus floc diameter for two hypothetical fractal groups ($n_f = 1.5$ and $n_f = 2.0$), both with a common primary particle diameter of 10 μm, using Eq. (4) (Fig. 11a). For $n_f = 1.5$, we fix $b_1 = 50$, and for $n_f = 2.0$, we vary b_1 from 10 to 100. As the shape factor increases with fractal dimension, the intersection diameter $d_{\text{intersect}}$ between the two groups becomes greater than the true primary particle diameter. The opposite holds if b_1 decreases with n_f.

We evaluate the calculated geometric intersection diameters ($d_{\text{intersect}}$) obtained under the standard assumption of a constant shape factor (Fig. 11b). While most measured floc diameters lie to the right of $d_{\text{intersect}}$, the lower tail of the observed floc size distribution inevitably extends to the left (i.e., some observed $d_f < d_{\text{intersect}}$). Because a floc physically cannot be smaller than its constituent primary particles ($d_f \geq d_p$), the true primary particle diameter must be strictly smaller than all measured floc diameters. Consequently, the assumption that $d_{\text{intersect}} = d_p$ produces a physical contradiction, requiring instead that the true primary particle size be securely anchored to the left of the constant-b_1 intersection ($d_p < d_{\text{intersect}}$).

This strict physical constraint implies a necessary relationship between b_1 and n_f. According to Eq. (7), the only way to maintain $d_p < d_{\text{intersect}}$ as n_f increases is if b_1 also increases. Such a trend is consistent with theoretical expectations and experimental observations, which suggest that denser, less permeable flocs (i.e., those with higher n_f) experience greater hydrodynamic drag, corresponding to higher values of b_1 (Strom & Keyvani, 2011). Importantly, this conclusion does not invalidate the n_f values inferred from the stratified $\log(w_s)$–$\log(d_f)$ slopes, because those slopes are independent of b_1; the shape factor enters only the intercept. The sensitivity of $d_{\text{intersect}}$ to b_1 therefore affects only the magnitude of the inferred primary particle diameter, not the fractal dimension itself nor the systematic variation of d_p across experimental conditions.

3.4 Reconciling Settling Dynamics with Variable Fractal Dimension

Having established that both n_f (Section 3.2) and d_p (Section 3.3) vary systematically with experimental conditions, we now test whether this structural variability explains the settling velocity discrepancies identified in Section 3.1. We formulate a modified model that retains the same equilibrium d_f prediction but evaluates $w_s \propto d_p^{3-n_f}\, d_f^{n_f-1}$ with the measured, condition-specific n_f and d_p rather than fixed values. As with the base model, we normalize the prediction by a single least-squares constant and plot it against the measured medians in Fig. 6c,d. However, both models share the identical $d_{f,\text{eq}}$ prediction and each contain exactly one free parameter, so any improvement in the modified model arises solely from incorporating the measured structural variability, not from additional fitting degrees of freedom.

As shown in Fig. 6c, the modified model successfully reproduces the non-monotonic peak in settling velocity with shear. At low to moderate shear ($u_* = 0.02$ to $0.03\ \text{m}\,\text{s}^{-1}$), the rapid densification of flocs (increasing n_f) outpaces the reduction in aggregate size, causing w_s to rise despite shrinking d_f. Only at high shear ($u_* = 0.04\ \text{m}\,\text{s}^{-1}$) does fragmentation dominate over densification, and w_s declines. Similarly, the modified model captures the steep monotonic decrease of w_s with organic matter (Fig. 6d): decreasing n_f and d_p with added OM reduce the effective solid fraction of the aggregates, overwhelming the minor increase in d_f.

Across all six experiments, the modified model reduces the root-mean-square error in predicted settling velocity by 34% relative to the base model. This improvement demonstrates that the discrepancies between the constant-n_f framework and our data are not random scatter but arise from systematic structural changes that the base model cannot represent. Properly predicting the settling dynamics of cohesive sediment therefore requires accounting for the dynamic variability of n_f and d_p with environmental forcing.

4 Discussion

A central finding of this research is the discrepancy between fractal dimensions derived from aggregated (i.e., pooled) datasets and those obtained from datasets stratified by 2D box-counting dimension. As illustrated by our conceptual model (Fig. 1) and supported by experimental results (Figs. 8 and 9), aggregating settling data from a floc population with inherent structural variability—meaning a range of true n_f values—can yield a bulk-fitted n_f that overrepresents certain structural subgroups while failing to capture the full diversity within the population. This approach can obscure underlying trends and produce fitted fractal-dimension values that are potentially misleading. The mechanism is analogous to Simpson's paradox: because distinct structural subpopulations occupy different regions of w_s–d_f space, the slope of a pooled regression is not a simple average of within-group slopes and may be dominated by the subpopulation that exerts the greatest leverage in log–log space.

4.1 Effect of Shear Velocity and Organic Matter

The settling velocity w_s governs the rate at which flocculated mud deposits from the water column and is therefore the quantity of most direct relevance to sediment transport predictions. For fractal flocs, $w_s \propto d_p^{3-n_f}\, d_f^{n_f-1}$ (Eq. (4)), which can be rewritten as $w_s \propto \phi\, d_f^2$ by defining the solid volume fraction $\phi = (d_f/d_p)^{n_f-3}$. In this form, settling is controlled jointly by floc size (d_f^2) and effective density (ϕ), where the latter encapsulates the combined influence of fractal dimension and primary particle size. This decomposition is useful because shear velocity and organic matter affect d_f and ϕ in distinct and sometimes opposing ways.

With increasing shear velocity, d_f decreases as stronger turbulence breaks apart aggregates (Fig. 6a), while n_f increases (Fig. 9a) and the inferred d_p coarsens (Fig. 10d), consistent with preferential retention of compact, shear-resistant structures (Winterwerp, 1998). The rising n_f and coarsening d_p increase ϕ, partially compensating for the shrinking d_f. Because these two contributions act in opposing directions on w_s, the observed settling response is non-monotonic (Fig. 6c). With increasing organic matter, n_f decreases (Fig. 9b) and d_p fines (Fig. 10h), consistent with organic binding promoting open, porous architectures built from smaller primary particles. Both effects reduce ϕ, and because d_f remains relatively stable across the OM gradient (Fig. 6b), w_s decreases monotonically (Fig. 6d).

The equilibrium floc diameter $d_{f,\mathrm{eq}}$, by contrast, is well predicted by the Nghiem et al. (2022) framework across all conditions (Fig. 6a,b), because n_f cancels in the aggregation–breakup balance and the predicted size depends only on η and θ. The settling velocity retains an explicit dependence on n_f and d_p, and a constant-n_f model cannot reproduce either the non-monotonic shear response or the organic-matter decline (Fig. 6c,d). Incorporating the measured structural variability (Section 3.4) reduces the root-mean-square error by 34% with no additional free parameters, confirming that these discrepancies reflect systematic structural changes rather than scatter. Although the exact quantitative relationships are likely system-specific, the general trends are broadly consistent with previous studies and underscore that accurate settling predictions require accounting for the variability of n_f and d_p with environmental forcing.

4.2 2D to 3D Conversion of Fractal Dimension

The relationship between the 2D box-counting dimension ($n_f^{(2\mathrm{D})}$) and the inferred 3D fractal dimension (n_f) is broadly consistent with models developed for synthetic computational aggregates (Fig. 9c). The simple Euclidean conversion, $n_f = n_f^{(2\mathrm{D})}+1$, systematically overestimates n_f across our measured range, whereas our data align more closely with the empirical conversion curves of Wang et al. (2022), based on computationally generated aggregates.

However, no universally valid mapping between 2D and 3D fractal dimensions exists for natural flocs. Most synthetic studies, including Wang et al. (2022), grow aggregates from monodisperse primary particles with fixed size, allowing simple scaling laws to be derived. Natural flocs, by contrast, contain a broad distribution of particle sizes, shapes, and compositions, adding heterogeneity in packing and porosity that is not captured by idealized models. Other approaches (Maggi & Winterwerp, 2004; Tang & Maggi, 2015) depend on their own specific assumptions and likewise cannot account for this variability. These differences introduce scatter and systematic offsets between natural data and synthetic conversions.

The close match between our results and the box-counting conversion of Wang et al. (2022) may therefore reflect structural similarities between our experimental flocs and their synthetic aggregates rather than a universally applicable rule. Nevertheless, the 2D box-counting dimension remains a practical stratification tool: by quantifying within-population variability in $n_f^{(2D)}$ and calibrating the 2D–3D relationship directly from experimental data, our approach provides context-sensitive inference of the hydrodynamically relevant 3D fractal dimension without relying on a fixed conversion law.

4.3 Constraining Primary Particle Diameter (d_p) and Shape Factor (b_1)

The intersection analysis yields $d_{\text{intersect}}$ distributions that agree in central tendency with the input sediment but are substantially narrower, peaking in the coarse-silt range (Fig. 10). This narrowing is expected on theoretical grounds. Nghiem et al. (2024) showed that the effective d_p in fractal scaling theory is better represented by a number-weighted moment of the size distribution rather than a volumetric median, because fractal aggregation conserves particle number rather than volume. The inferred $d_{\text{intersect}}$ is therefore expected to deviate from the bulk median grain size in a manner that depends on the weighting imposed by the aggregation process. Additionally, the finest clays may remain dispersed under the experimental freshwater chemistry, while the coarsest silica grains may be too heavy to be incorporated into flocs—both effects narrow the effective d_p distribution relative to the input.

The intersection method assumes a constant b_1 across n_f classes (Eq. (8)); when b_1 varies with floc structure, $d_{\text{intersect}}$ deviates from the true d_p (Eq. (7)). Our analysis (Fig. 11) shows that physically plausible intersections require b_1 to increase with n_f, consistent with the expectation that denser, less permeable flocs experience greater form drag (Strom & Keyvani, 2011). However, this sensitivity affects only the absolute magnitude of $d_{\text{intersect}}$. The n_f values inferred from stratified slopes are independent of b_1, which enters only the intercept, and the relative trends in $d_{\text{intersect}}$ across experimental conditions are preserved because the same $b_1(n_f)$ relationship applies uniformly to all experiments. Independently constraining $b_1(n_f)$—for example through direct drag measurements on size-selected flocs—would allow the intersection estimate to be converted into a true d_p.

Floc permeability, not considered here, may further complicate the inference. Permeability reduces the effective drag coefficient and may itself depend on d_f and d_p; if it varies systematically with floc size, assuming it to be constant can bias both the inferred slope and therefore n_f (Nghiem et al., 2024). Quantifying the joint dependence of b_1 and permeability on floc structure remains an important target for future work.

4.4 Implications

The 3D fractal dimension of mud flocs, n_f, is a key parameter in morphodynamic models of fine-grained sediment transport. Historically, modelers have generally assumed

$n_f \approx 2$ (Kranenburg, 1994; Son & Hsu, 2008; Winterwerp, 1998; J. C. Winterwerp & Van Kesteren, 2004). These studies inferred the fractal dimension by aggregating settling velocity data from entire floc populations. When we apply this same classical aggregation technique to our own experimental data, we derive a similar fractal dimension of approximately 2.0. This consistency confirms that our experimental flocs are physically comparable to those observed in previous work. However, this apparent agreement masks a significant underlying complexity.

By stratifying flocs according to their 2D box-counting dimension, our method reveals systematic trends that are invisible to conventional data aggregation. We find that the median n_f increases with shear velocity (from 1.77 to 2.12) and decreases with increasing organic matter content (from 1.84 to 1.63). When the stratified n_f and d_p values are fed back into the settling velocity equation (Section 3.4), the modified model reduces the prediction error by 34% relative to the constant-n_f base model, providing direct evidence that this structural variability is dynamically significant. In contrast, the aggregated analysis compresses the dynamic range by roughly a factor of five, yielding median values that vary by only ~0.06 across the same experimental conditions compared with ~0.2–0.35 for the stratified approach. The aggregated estimates also regress toward a population-average $n_f \approx 2$, reflecting Simpson's paradox: fitting a single slope to a structurally mixed population of loose and dense flocs yields an intermediate value that neither represents the dominant subgroup nor captures the underlying environmental trend. The result is that relying on a single, representative fractal dimension near 2.0, as is typical in existing models, both overstates floc compactness under conditions that produce porous flocs and obscures the sensitivity of n_f to shear and organic matter.

This finding has important implications for sediment transport prediction. Using the Stokes' law–based formulation for floc settling velocity, $w_s \propto d_p^{3-n_f}\, d_f^{n_f-1}$, lowering n_f from 2.0 to 1.7 for a typical 400 µm floc (with 4 µm primary particles, a representative value from the literature) produces a several-fold decrease in settling speed, which directly translates to a several-fold increase in transport distance. In many transport formulations the deposition flux scales as $F_{\mathrm{dep}} \sim w_s\, C$ (where C is the suspended sediment concentration) (Amoudry, 2008), so lowering n_f—and thus w_s—reduces modeled deposition rates and increases downstream export for the same supply. Flocs that might be predicted to settle out over a given distance would travel several times farther under this revised, more porous structural model. As a result, mud flocs in natural environments may settle more slowly and be transported substantially farther than predicted by existing models that rely on aggregated fractal dimensions. The efficiency of mud retention, at least in the freshwater environments considered here, may be considerably lower than currently assumed, with significant consequences for sediment budgets, contaminant dispersal, and carbon burial in floodplains and deltas.

5 Conclusion

This study demonstrates that accounting for heterogeneity within floc populations is essential for accurate sediment transport modeling. Aggregating settling data across flocs with varying fractal dimensions can produce misleading results due to Simpson's paradox: trends present within structural subgroups are lost or distorted when data are pooled. By stratifying flocs based on their 2D box-counting fractal dimension, we resolve structural differences that conventional bulk analysis obscures.

The stratified approach reveals that n_f increases with shear velocity and decreases with organic matter content, while the inferred d_p co-varies with these same conditions—coarsening under stronger shear and fining under organic binding. The concurrent variation of n_f, d_p, and d_f introduces competing effects on $w_s \propto d_p^{3-n_f}\, d_f^{n_f-1}$, producing non-monotonic settling responses that constant-n_f models cannot capture. Incorporating the measured structural variability into the settling velocity prediction reduces the root-mean-square error by 34% relative to the constant-n_f base model, with no additional free parameters. The framework also yields an empirical 2D–3D fractal dimension relationship that matches existing synthetic-aggregate conversions and shows that b_1 must increase with n_f to maintain physical consistency.

These results indicate that the conventional assumption of $n_f \approx 2$ overestimates floc compactness and settling rates. Reducing n_f to the observed range of 1.6–2.1 can decrease predicted w_s by several-fold for typical floc sizes, implying that mud retention in freshwater systems may be considerably lower than current models suggest. We recommend adopting a lower default n_f and, where feasible, representing the fractal dimension as a distribution. While no single value captures the full complexity of natural flocculation, shifting away from the classical range is a necessary step toward more accurate predictions of fine sediment transport, deposition, and downstream sediment budgets.

Open Research Section

All code and analysis workflows supporting the results and conclusions of this study are openly available at `https://github.com/braydennoh/FlocTrack`. The repository includes scripts for image analysis, fractal dimension calculation, and all processing steps described in this manuscript. For further questions or access to additional datasets, please contact the corresponding author at braydennoh@fas.harvard.edu.

Inclusion in Global Research Statement

This research did not involve collaborations with local communities or cross-regional partnerships requiring additional disclosure. All research activities were conducted in accordance with institutional and international ethical standards.

Acknowledgments

BN acknowledges support from Caltech SURF, the Arthur R. Adams SURF Fellowship, and the Saul and Joan Cogen Memorial SURF Fellowship at Caltech. MPL acknowledges support from NASA FINESST Grant 80NSSC20K1645, the NASA Delta-X project funded by the Science Mission Directorate's Earth Science Division through the Earth Venture Suborbital-3 Program NNH17ZDA001N-EVS3, and the National Science Foundation Geomorphology and Land-use Dynamics Grant No. 2136991.

References

Amoudry, L. (2008). A review on coastal sediment transport modelling.

Bellouti, M., Alves, M., Novais, J., & Mota, M. (1997). Flocs vs granules: differentiation by fractal dimension. *Water Research*, *31*(5), 1227–1231.

Bianchi, T. S., Mayer, L. M., Amaral, J. H., Arndt, S., Galy, V., Kemp, D. B., ... Regnier, P. (2024). Anthropogenic impacts on mud and organic carbon cycling. *Nature Geoscience*, *17*(4), 287–297.

Bird, R. (2006). Transport phenomena wiley. *Transport phenomena Wiley. 2nd edition.*

Blair, N. E., & Aller, R. C. (2012). The fate of terrestrial organic carbon in the marine environment. *Annual review of marine science*, *4*(1), 401–423.

Camp, T. R., & Stein, P. C. (1943). Velocity gradients and internal work in fluid motion. *Journal of the Boston Society of Civil Engineers*, *30*, 219–237.

Cheng, Q. (1995). The perimeter-area fractal model and its application to geology. *Mathematical Geology*, *27*, 69–82.

Daily, J. W., & Nece, R. E. (1960). Chamber dimension effects on induced flow and frictional resistance of enclosed rotating disks. *Journal of Basic Engineering*, *82*(1), 217–230.

Douglas, M. M., Miller, K. L., & Lamb, M. P. (2025). Mud cohesion governs unvegetated meander migration rates and deposit architecture. *Geological Society of America Bulletin*, *137*(1-2), 522–540.

Droppo, Leppard, G., Flannigan, D., & Liss, S. (1997). The freshwater floc: a functional relationship of water and organic and inorganic floc constituents affecting suspended sediment properties. In *The interactions between sediments and water: Proceedings of the 7th international symposium, baveno, italy 22–25 september 1996* (pp. 43–53).

Esposito, C. R., Shen, Z., Törnqvist, T. E., Marshak, J., & White, C. (2017). Efficient retention of mud drives land building on the mississippi delta plain. *Earth Surface Dynamics*, *5*(3), 387–397.

Family, F., & Landau, D. P. (2012). *Kinetics of aggregation and gelation*. Elsevier.

Ferguson, R., & Church, M. (2004). A simple universal equation for grain settling velocity. *Journal of sedimentary Research*, *74*(6), 933–937.

Foroutan-pour, K., Dutilleul, P., & Smith, D. L. (1999). Advances in the implementation of the box-counting method of fractal dimension estimation. *Applied mathematics and computation*, *105*(2-3), 195–210.

Frame, M., Mandelbrot, B., & Neger, N. (2025). *Fractal geometry.* `https://gauss.math.yale.edu/fractals/`. (Yale University. Accessed: 2025-05-22)

Healy, T., Wang, Y., & Healy, J.-A. (2002). *Muddy coasts of the world: processes, deposits and function* (Vol. 4). Elsevier.

Jarvis, P., Jefferson, B., & Parsons, S. A. (2005). Measuring floc structural characteristics. *Reviews in Environmental Science and Bio/Technology*, *4*, 1–18.

Johnson, C. P., Li, X., & Logan, B. E. (1996). Settling velocities of fractal aggregates. *Environmental science & technology*, *30*(6), 1911–1918.

Julien, P. Y. (2018). *River mechanics*. Cambridge University Press.

Jullien, R. (1987). Aggregation phenomena and fractal aggregates. *Contemporary Physics*, *28*(5), 477–493.

Kemp, G. P., Day, J. W., Rogers, J. D., Giosan, L., & Peyronnin, N. (2016). Enhancing mud supply from the lower missouri river to the mississippi river delta usa: Dam bypassing and coastal restoration. *Estuarine, Coastal and Shelf Science*, *183*, 304–313.

Khelifa, A., & Hill, P. S. (2006). Models for effective density and settling velocity of flocs. *Journal of Hydraulic Research*, *44*(3), 390–401.

Kranenburg, C. (1994). The fractal structure of cohesive sediment aggregates. *Estuarine, Coastal and Shelf Science*, *39*(5), 451–460.

Lamb, M. P., de Leeuw, J., Fischer, W. W., Moodie, A. J., Venditti, J. G., Nittrouer, J. A., ... Parker, G. (2020). Mud in rivers transported as flocculated and suspended bed material. *Nature Geoscience*, *13*(8), 566–570.

Lee, C., & Kramer, T. A. (2004). Prediction of three-dimensional fractal dimensions using the two-dimensional properties of fractal aggregates. *Advances in colloid and Interface Science*, *112*(1-3), 49–57.

Li, D. H., & Ganczarczyk, J. (1989). Fractal geometry of particle aggregates generated in water and wastewater treatment processes. *Environmental science & technology*, *23*(11), 1385–1389.

Liberzon, A., Lasagna, D., Aubert, M., Bachant, P., Mahmoodtabar, E., Käufer, T., ... others (2021). Openpiv/openpiv-python: Openpiv-python v0. 22.3. *Zenodo*.

Liss, S. N., Droppo, I. G., Flannigan, D. T., & Leppard, G. G. (1996). Floc architecture in wastewater and natural riverine systems. *Environmental science & technology*, *30*(2), 680–686.

Maggi, F. (2007). Variable fractal dimension: A major control for floc structure and flocculation kinematics of suspended cohesive sediment. *Journal of Geophysical Research: Oceans*, *112*(C7).

Maggi, F., & Winterwerp, J. (2004). Method for computing the three-dimensional capacity dimension from two-dimensional projections of fractal aggregates. *Physical Review E*, *69*(1), 011405.

Mandelbrot, B. (1967). How long is the coast of britain? statistical self-similarity and fractional dimension. *science*, *156*(3775), 636–638.

Mason, J., & Mohrig, D. (2019). Differential bank migration and the maintenance of channel width in meandering river bends. *Geology*, *47*(12), 1136–1140.

Meakin, P. (1992, oct 1). Aggregation kinetics. *Physica Scripta*, *46*(4), 295–331. doi: 10.1088/0031-8949/46/4/002

Nghiem, J. A. (2025). *Flocculation and transport of mud in rivers and deltas* (Unpublished doctoral dissertation). California Institute of Technology.

Nghiem, J. A., Fischer, W. W., Li, G. K., & Lamb, M. P. (2022). A mechanistic model for mud flocculation in freshwater rivers. *Journal of Geophysical Research: Earth Surface*, *127*(5), e2021JF006392.

Nghiem, J. A., Li, G. K., Harringmeyer, J. P., Salter, G., Fichot, C. G., Cortese, L., & Lamb, M. P. (2024). Testing floc settling velocity models in rivers and freshwater wetlands. *Earth Surface Dynamics*, *12*(6), 1267–1294.

Nghiem, J. A., Li, G. K., Zepeda-Rosales, M., Li, Y., & Lamb, M. P. (2026). Flocculated silt, not clay, dominates river suspended sediment load. *Geophysical Research Letters*, *53*(2), e2025GL118655.

Nicholas, A., & Walling, D. (1996). The significance of particle aggregation in the overbank deposition of suspended sediment on river floodplains. *Journal of Hydrology*, *186*(1-4), 275–293.

Osborn, R., Dillon, B., Tran, D., Abolfazli, E., Dunne, K. B., Nittrouer, J. A., & Strom, K. (2021). Flocarazi: An in-situ, image-based profiling instrument for sizing solid and flocculated suspended sediment. *Journal of Geophysical Research: Earth Surface*, *126*(11), e2021JF006210.

Pertuz, S., Puig, D., & Garcia, M. A. (2013). Analysis of focus measure operators for shape-from-focus. *Pattern Recognition*, *46*(5), 1415–1432.

Pope, S. B. (2000). *Turbulent flows*. Cambridge University Press.

Schlichting, H., & Gersten, K. (2000). *Boundary-layer theory*. Springer.

Simpson, E. H. (1951). The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, *13*(2), 238–241.

Smith, S. J., & Friedrichs, C. T. (2015). Image processing methods for in situ estimation of cohesive sediment floc size, settling velocity, and density. *Limnology and Oceanography: Methods*, *13*(5), 250–264.

Son, M., & Hsu, T.-J. (2008). Flocculation model of cohesive sediment using variable fractal dimension. *Environmental Fluid Mechanics*, *8*(1), 55–71.

Spencer, K., Wheatland, J., Carr, S., Manning, A., Bushby, A., Gu, C., ... Lawrence, T. (2022). Quantification of 3-dimensional structure and properties of flocculated natural suspended sediment. *Water Research*, *222*, 118835.

Stokes, G. G., et al. (1851). On the effect of the internal friction of fluids on the motion of pendulums.

Strom, K., & Keyvani, A. (2011). An explicit full-range settling velocity equation for mud flocs. *Journal of Sedimentary Research*, *81*(12), 921–934.

Tang, F. H., & Maggi, F. (2015). Reconstructing the fractal dimension of granular aggregates from light intensity spectra. *Soft matter*, *11*(47), 9150–9159.

Wang, R., Singh, A. K., Kolan, S. R., & Tsotsas, E. (2022). Fractal analysis of aggregates: Correlation between the 2d and 3d box-counting fractal dimension and power law fractal dimension. *Chaos, Solitons & Fractals*, *160*, 112246.

Winterwerp. (1998). A simple model for turbulence induced flocculation of cohesive sediment. *Journal of hydraulic research*, *36*(3), 309–326.

Winterwerp. (2002). On the flocculation and settling velocity of estuarine mud. *Continental shelf research*, *22*(9), 1339–1360.

Winterwerp, J., Manning, A., Martens, C., De Mulder, T., & Vanlede, J. (2006). A heuristic formula for turbulence-induced flocculation of cohesive sediment. *Estuarine, coastal and shelf Science*, *68*(1-2), 195–207.

Winterwerp, J. C., & Van Kesteren, W. G. (2004). *Introduction to the physics of cohesive sediment dynamics in the marine environment* (Vol. 56). Elsevier.

Zeichner, S. S., Nghiem, J., Lamb, M. P., Takashima, N., de Leeuw, J., Ganti, V., & Fischer, W. W. (2021). Early plant organics increased global terrestrial mud deposition through enhanced flocculation. *Science*, *371*(6528), 526–529.